

Restaurant Management System

For

Palladium Restaurant & Bar

Project Report



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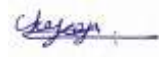
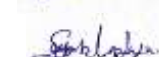
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Declaration

This project report is our original work, and the content is not plagiarized from any other resource. References for all the content taken from external resources are correctly cited. To the best of our knowledge, this report does not contain any material published or written by third parties, except as acknowledged in the text.

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Abstract

This report describes a restaurant management system that is designed to streamline the ordering and inventory management process for a restaurant. The system allows customers to place orders online, on-call, or by visiting the restaurant for take-away, dine-in, or delivery. For dine-in orders, a particular waiter and table are assigned to the order, while a particular delivery person is assigned for delivery orders, and payment is made to the delivery person responsible for the order.

The system features a built-in inventory management system that allows the inventory manager to monitor the available stocks of both the restaurant and bar and receive notifications when the stocks reach the reorder level. Order management is performed using a printed token, with items from the restaurant printed in the Kitchen Order Ticket (KOT) and bar items in Bar Order Ticket (BOT). The kitchen receives the KOT and prepares the order accordingly.

Once the order is prepared, the KOT and BOT are matched with the invoice numbers and handed over to the cashier, then the cashier hands over the printed bill. This system is designed to increase efficiency in the ordering process, reduce errors, and improve the overall customer experience. By integrating online ordering and inventory management, the restaurant can optimize its operations, reduce waste, and increase profits.

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We, the team members of ITP_KDY_2023_WD_1 Restaurant management system project, would like to express our sincere appreciation to all those who have contributed to the success of this project.

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Sincerely,

Team ITP_KDY_2023_WD_1

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List of abbreviations

Abbreviation	Definition
MERN	MongoDB, Express, React, Node
KOT	Kitchen-Order-Ticket
BOT	Bar-Order-Ticket
OTP	One-time password
FAQ	Frequently asked questions
DB	Database
UI	User interface

1 Introduction

1.1 Background

In today's fast-paced world, technology has made its way into almost every aspect of our lives. The restaurant industry is no exception, and restaurant management systems have become an essential tool for managing the operations of a restaurant. These systems allow restaurant owners and managers to streamline their business operations, automate processes, and improve the overall efficiency of their restaurant.

The purpose of this report is to describe a restaurant management system that we have designed, which allows customers to place orders online, on-call, or by visiting for Take-away, Dine-in, or Delivery. This system includes a built-in inventory management system that helps the inventory manager to keep track of the available stocks of both the restaurant and bar, and get notified when the stocks reach the re-order level. The system also includes an order management module that uses printed tokens to ensure accurate order fulfillment.

We, a team of 2nd year undergraduates of SLIIT campus following BSc(Hons) in Information Technology, designed this restaurant management system as part of our coursework to solve the problem of manual order taking and management at a local restaurant. Our goal was to design a system that not only improves the customer experience but also helps the restaurant staff to manage orders, inventory, and deliveries efficiently while satisfying our client's requirements. In this report, we will describe the key features of our system, including the system architecture, modules, and functionalities. We will also discuss the challenges we faced during the development process and the solutions we implemented to overcome them.

1.2 Problem and motivation

The restaurant industry is a highly competitive market with tight profit margins. To be successful, restaurant owners and managers must focus on maximizing efficiency and providing excellent customer service. However, manual order taking and management can be time-consuming, error-prone, and may lead to a poor customer experience. In addition, manually managing inventory, deliveries, and payments can be a daunting task for restaurant staff.

Our team recognized these challenges faced by our client and other local restaurants and decided to design a restaurant management system that addresses these issues. Our motivation was to provide a solution that streamlines restaurant operations, reduces errors, and improves the overall customer experience. We aimed to design a system that is user-friendly, easy to use, and affordable for small and medium-sized restaurants.

We conducted a thorough analysis of the existing restaurant management systems in the market and the system currently used by our client and found that most of them are expensive, complex, and difficult to use. In addition, many of these systems are not customizable, making them unsuitable for the specific needs of small and medium-sized restaurants. Therefore, we saw an opportunity to design a system that is tailored to the needs of local restaurants, affordable, and easy to use.

Some of the problems we identified by our client were:

Our client currently uses a desktop application. But according to our client, that software does not address his requirements.

- The software has failed to produce daily reports of sales and inventory.
- Our client has noticed several frauds and wastage in inventory.
- Currently running a hybrid system where there isn't any online ordering system unless the customer could place orders on-call or arriving to the restaurant.
- A delivery system doesn't exist.
- A separate PC is allocated as the server where if the server downs the whole system runs manually.
- Not very user friendly and need additional training in using the software.
- Due to the drastic growth in technology which in turn automates the entire corporate world therefore, our client too intends to develop and adapt to the changes.

In this report, we will describe the key features of our restaurant management system, including its modules, functionalities, and architecture. We will also discuss the advantages of our system over the existing solutions in the market and provide a detailed analysis of its benefits. Our aim is to demonstrate how our system can help local restaurants improve their efficiency, reduce errors, and enhance the overall customer experience by addressing the requirements of our client.

1.3 Literature review

With the advancement of technology currently majority of the businesses including restaurants do Business to Consumer interactions online. Due to this factor, designing and developing a proper web application is vital to attract more customers to a business. A well-designed website with high usability has been found to positively influence visitor retention (revisit rates) and purchasing behavior. [1]

The required system is a combination of an online ordering system as well as a restaurant management system to manage orders, inventory, and staff. Therefore, already existing similar systems were analyzed to get a better idea about the development process of the required web application.

Pizza Hut is multinational restaurant chain which is popular around the world. In the local Pizza Hut web application, you can place orders online. Both takeaway and delivery options are available on the website. A simple menu including all the dishes/items are available and you can easily add items to your cart or remove them as you wish. The specialty in the menu is, you can customize any item to your liking as the restaurant provides many toppings/crusts, etc. Once you add the necessary items to your cart you can checkout and select the method of payment and place your order. Additional features like selecting the pick-up location are also available on the website. Since Pizza Hut has many stores around the country the website has been made to allow customers from different areas to place orders online. But the client has only one restaurant and the target audience is limited. Therefore, features like selecting and locating the nearest store are not needed for the required system. Also, since the restaurant's menu is not customizable when placing orders, the customizable menu feature is also not necessary for the required system. Additionally, not being able to get real-time customer support, inability add favorite items are few of the drawbacks of this website. [2]

Upserve is a restaurant management platform which allows restaurants to manage their business. In the system, you can manage the restaurant's inventory, menu, orders, and staff. [3] With the availability of inventory management, you can increase the productivity and transparency within the restaurant by reducing manual stock counting and other operations. Also, the ability to add, modify and remove menu items quickly within a short amount of time is a major benefit of this system. Along with these features there are some unwanted components of the system like staff management where work hours of each staff member are calculated, and employee reports are made daily. Since the client has specifically mentioned about adding only waiter and delivery management, the staff management option used in the Upserve platform is additional and unnecessary. And, since the restaurant does not handle online payment methods, the payment handling option is also not a requirement. Briefly, the client requires a less complex system when compared to Upserve, which has a lot of advanced features which suit bigger restaurants, having multiple stores and bigger staff to manage.

In conclusion, to fulfil the client's requirement, a restaurant management system along with an online ordering system needs to be implemented together. Therefore, a new system which has minimum complexities and user friendly must be built.

1.4 Aim and Objectives

1.4.1 Aim

This system is primarily developed to perform the day-to-day functionalities conveniently and with ease for both customers and the administration with good usability and reliability with 100% data security. Helps the restaurant manager to manage the restaurant more effectively and efficiently by computerizing meal ordering, billing, and inventory control.

1.4.2 Objectives

To achieve the aim of this project, we have set the following objectives:

- To design a software system that provides daily reports of sales and inventory, allowing our client to better track their business performance and identify any discrepancies or frauds.
- To develop an inventory management module that reduces the wastage of inventory and ensures efficient utilization of resources.
- To design and implement an online ordering system that enables customers to place orders from anywhere, providing a convenient and efficient ordering experience.
- To develop a delivery management system that streamlines the delivery process and ensures accurate delivery tracking.
- To design a server-client architecture that ensures the system remains operational even if the server is down, minimizing downtime and maximizing system availability.
- To create a user-friendly interface that requires minimal training for restaurant staff and enables efficient management of sales, inventory, and orders.
- To ensure the system is scalable, adaptable, and customizable to meet the future needs of our client and other small and medium-sized restaurants.

By achieving these objectives, we aim to provide a comprehensive restaurant management system that streamlines business operations, reduces errors and wastage, improves the overall customer experience, and enables our client to adapt to the changing business landscape.

1.5 Solution overview

The new restaurant management system has been designed and developed as a comprehensive solution that addresses the challenges faced by our client. The system includes the following features:

- **Sales and Inventory Management:**
The software system provides daily reports of sales and inventory, allowing our client to better track their business performance. The inventory management module is designed to reduce the wastage of inventory and ensure efficient utilization of resources.
- **Online Ordering System:**
The online ordering system enables customers to place orders from anywhere, providing a convenient and efficient ordering experience. Customers can place their orders online, on-call, or by visiting for Take-away, Dine-in, or Delivery.
- **Delivery Management System:**
The delivery management system streamlines the delivery process and ensures accurate delivery is made on time. A particular delivery person will be assigned to each delivery order, and the payment will be done to that delivery person who is responsible for that order.
- **Server-Client Architecture:**
The new system is designed as a server-client architecture, ensuring the system remains operational even if the server is down, minimizing downtime, and maximizing system availability.
- **User-Friendly Interface:**
The system includes a user-friendly interface that requires minimal training for restaurant staff and enables efficient management of sales, inventory, and orders. The order management is performed with the use of a printed token, and the kitchen and bar receive the KOT and BOT respectively to prepare the order accordingly.
- **Inventory Management System:**
The new system is scalable, adaptable, and customizable to meet the future needs of our client and other small and medium-sized restaurants. The built-in inventory management system allows the inventory manager to get a clear understanding of the available stocks of both the restaurant and bar and get notified when the stocks reach the reorder level.

Overall, the new restaurant management system provides a comprehensive solution that streamlines business operations, reduces errors and wastage, improves the overall customer experience, and enables our client to adapt to the changing business landscape.

1.6 Product Scope

User management

The customer management function in the restaurant management system is designed to provide a seamless experience for customers. The system allows customers to log in using their registered email and password. Additionally, customers have the option to log in using their Google accounts, leveraging Google authentication for a convenient and secure login process.

If a customer forgets their password, a password reset feature is available. By providing their email address, customers can request a password reset OTP. Upon entering the OTP, they are prompted to enter a new password and verify their identity with a four-digit code.

The customer management process encompasses various tasks, including customer profile creation, deletion, updating, and retrieval. The system also allows for easy profile updates, enabling customers to modify individual data sets within their profiles as needed.

Maintaining accurate and up-to-date customer records is vital for system administrators and business owners. The system ensures that client information is readily available and easily managed, facilitating effective communication and personalized service. Moreover, the option to log in using Google accounts enhances security and offers a seamless login experience for customers.

Food management

The food management module of the restaurant management system provides various essential functions for efficient management of dishes and their related details. These functions are designed to streamline the process of adding, editing, and removing dish information, as well as providing comprehensive reports on dish items. The following functions are available within the food management module:

Add/Remove or Edit Dish Details: This function allows authorized personnel to add new dishes to the system, specifying relevant details such as dish name, category, price, and ingredients used. Additionally, the system provides the capability to edit or remove existing dish information as needed, ensuring the menu remains up to date.

Attach Images of Dishes: To enhance visual appeal and assist customers in making informed choices, the system allows images of the dishes to be attached. This feature enables staff to upload high-quality images that accurately represent the appearance and presentation of each dish. AWS S3 bucket is used to facilitate this service.

Manage Ingredients and Inventory Stock Items: In order to maintain accurate inventory records and ensure availability of ingredients, the system allows for the management of ingredients used in each dish. Staff can associate the necessary ingredients with the corresponding dishes, utilizing inventory stock items. This helps in monitoring ingredient usage, tracking inventory levels, and ensuring proper stock management.

Browse Detailed View of Dish Items: The system provides a user-friendly interface for browsing and accessing detailed information about each dish. This feature allows users to view comprehensive details such as dish ingredients and amounts of ingredients used in each dish.

Generate Suitable Dish Reports: To facilitate effective decision-making and analysis, the system offers the capability to generate comprehensive reports about dish items. These reports can include

various parameters such as dish popularity, sales trends, inventory usage, and profitability. Such reports assist management in assessing the performance of different dishes, identifying customer preferences, and optimizing the menu selection.

The food management functions described above contribute to efficient dish management, accurate inventory control, and improved customer satisfaction. They enable staff to maintain an up-to-date and visually appealing menu, manage ingredient usage effectively, and generate valuable insights through comprehensive dish reports.

Cart management

The cart management feature allows customers to add and remove items they wish to purchase easily. Customers can add an item to their cart by clicking the "Add to cart" button under each item in the menu. The added items will be listed in the cart with their respective prices and quantities, and customers can adjust the quantity or remove items from the cart as needed. The cart can also be cleared at once. Once the customer is ready to purchase the items, they can proceed to checkout where the order details will be displayed. To enhance user experience, the shopping cart feature must be fast and responsive, ensuring customers can add, remove and edit items quickly and efficiently. It should also be easy to use for customers with clear instructions and user-friendly designs. In addition, it must be compatible with any device and web browser, scalable to handle large traffic during peak hours, and easy to maintain and edit with clear documentation and code base.

Menu category management

The menu category management feature allows the administrator to add categories to sort the menu food items. The category name can be edited, and the administrator can remove a category that has been added. A category ID is auto-generated, and menu items are sorted and displayed according to categories by retrieving data added by food management. This feature should be fast and responsive to ensure users can navigate through the menu quickly and efficiently. It should also be interactive with attractive interfaces, reliable and stable with minimal downtime, easy to use by customers and staff, compatible with any device and web browser, and easy to maintain and update with clear documentation and code base. By implementing this feature, the system can effectively manage the menu items, making it easier for customers to find what they are looking for and enhancing their overall experience.

Order management

Order Management is a crucial aspect of the system, which enables cashiers to add, manage, and track customer orders efficiently. Using this feature, cashiers can add a new order to the system and print the bill, including the KOT and the BOT. This module is also responsible of printing the KOT and BOT and sending it to the kitchen or the bar manually through a waiter. In case of a dining order, the cashier can assign a waiter to the order, while for delivery orders, the cashier can assign a driver. The system also allows cashiers to save customer details for future orders, making it easy to process orders quickly. Additionally, the invoices can be managed easily by changing or removing the entered items' details, and managers can view the previous order details.

Dashboard management

Dashboard Management is another crucial feature that enables managers to view the income, expenses, and profit of the daily and monthly operations of the system. The dashboard provides a summary of customers, orders, foods, and inventory stock, and a bar chart depicts the number of orders placed according to dining, delivery, and takeaway. Furthermore, the system displays the top 10 daily sales, making it easy for managers to identify the popular dishes.

FAQ management

The FAQ Management feature allows the administrator to add, edit, and delete FAQs under different categories. Each FAQ can have a question and an appropriate solution or answer. The administrator can also update the FAQs at any time to reflect changes in the system.

From the customer's perspective, they can access the FAQs page and browse through the questions and solutions provided by the administrator. The FAQs are categorized for easier navigation, and if a customer has a specific problem or question, they can use the search bar to quickly find the relevant information.

In addition to browsing the FAQs, customers can also initiate a real-time chat with the administrator. This chat feature allows customers to send messages directly to the administrator to ask questions or get help with any issues they are facing. The administrator can then respond to the customer's messages in real-time, providing prompt and helpful customer support.

Overall, the FAQ Management feature is designed to provide customers with easy access to information and support, improving their overall experience with the system. The feature is user-friendly and easy to maintain, ensuring that the FAQs are always up-to-date and relevant to customers' needs.

Restaurant Inventory management

Restaurant inventory management is a crucial aspect of any restaurant as it ensures that the right amount of ingredients and materials are available for preparing dishes. The inventory manager has the ability to add, remove and update inventory items. The administrator can access a detailed report on the stock, re-order level and availability of items in the inventory. When the re-order level is reached, the system automatically notifies the inventory manager to refill the stock.

As orders are placed, the ingredients used by the orders are reduced from the inventory in real-time. This helps to keep track of the inventory levels and ensures that there are no shortages. The system also automatically updates the stock when an item has reached its expiry date.

One of the key features of the inventory management system is the ability to analyze the quantity of items purchased in the previous 12 months through the generated bar chart. This information can be used to forecast the demand for certain ingredients and materials in the future, which helps to optimize the inventory levels and reduce wastage.

Overall, the inventory management system helps to ensure that the restaurant always has the right amount of ingredients and materials for preparing dishes, reduces wastage and helps to optimize costs.

Bar Inventory management

The bar inventory system is a crucial component of the overall inventory management system of a restaurant. It enables the restaurant to keep track of the inventory levels of all bar items, such as liquor, beer, and wine. The inventory manager is responsible for adding new bar items to the inventory, removing existing items, and updating the inventory as required.

The inventory manager maintains detailed records of each bar item, including the item's ID, name, quantity, price, and reorder level. When new bar items become available on the market, they are added to the inventory system. The inventory manager sets a minimum quantity level for each item currently in stock. When a bar item is sold, the system decrements the quantity of that item in the inventory.

If the quantity of a bar item falls below the set minimum quantity level, the system automatically notifies the inventory manager to make a supply order. This ensures that the bar items are always available to customers and that the inventory levels are maintained at optimal levels.

The inventory system also allows to remove bar items that have expired, ensuring that only fresh and high-quality items are available to customers. The system displays all available bar items to the customers, enabling them to make informed decisions about their drinks.

Overall, an efficient bar inventory system is critical for ensuring the smooth operation of a restaurant's bar, providing customers with high-quality and fresh drinks, and optimizing inventory levels to reduce waste and costs.

Waiter management

Waiter management is an essential part of restaurant operations. The primary objective is to provide customers with a high level of satisfaction. The admin is responsible for managing waiters in the system. The admin can add new waiters by entering their ID, name, email, address, phone number, and password into the system. These details are stored in the database for future reference.

The admin has the capability to update the waiter details if there are any changes that need to be made. For example, if a waiter changes their phone number or address, the admin can easily update their details in the system. The admin also has the authority to remove waiters from the system if necessary.

In summary, the waiter management system is a critical component of restaurant operations. It allows the admin to manage waiters efficiently.

Notifications management

The system has a notification feature that alerts the admin when certain events occur. There are three types of notifications that the admin can receive. The first notification is sent when the restaurant's stock level falls below a specific level. This is important because it enables the admin to take necessary steps to restock the inventory. The second notification is sent when someone makes an online order. This notification allows the admin to process the order promptly. The third notification is sent when a customer sends a pending incoming chat message. This notification enables the admin to respond to the customer's request as soon as possible.

This feature ensures that the admin can respond quickly to critical events in the restaurant.

Driver management

The system provides a platform for drivers to access and manage pending delivery orders. They can see a list of orders that are awaiting pickup. However, they are limited to choosing only one order at a time, ensuring a focused and organized workflow.

Once a driver selects an order, they are required to enter a start code to initiate the trip. This code serves as verification and ensures that the driver has officially started the delivery process. Additionally, the system automatically sends the order details and an end code to the customer, notifying them about the driver's engagement.

After the delivery is completed, the driver enters an end code into the system, indicating that the trip has concluded. This step helps to maintain accurate records and streamline the delivery process.

Both drivers and administrative personnel have access to a map interface where they can view various details such as the route, distance, and estimated delivery time. This feature allows for efficient navigation and real-time tracking of orders, enhancing coordination and customer satisfaction.

Furthermore, the system offers drivers and management personnel the ability to review their order completion summaries. This feature provides valuable insights into order history, including statistics on completed deliveries, which can help improve overall performance and identify areas for optimization.

Lastly, the track order panel serves as an interface to monitor the states of all delivery orders. This centralized feature allows management persons to track the progress of individual orders, ensuring transparency and efficient management of the delivery process.

1.7 The structure of the report

1.7.1 Methodology:

The methodology section will provide a detailed description of the research approach and development process used to create the restaurant management system. This section will cover the following topics:

- Research methods used to identify client requirements
- Software development methodology
- Development tools and technologies used
- Description of the system architecture and design
- Database schema and data model

1.7.2 Testing and Evaluation:

The testing and evaluation section will describe the various testing methodologies used to ensure that the restaurant management system is functional, efficient, and user-friendly. This section will cover the following topics:

- Testing methods used
- Types of testing, such as unit testing, integration testing, system testing, and acceptance testing
- Results of the testing phase
- Evaluation of the system's usability and user satisfaction
- Issues encountered during testing and their resolution

1.7.3 Conclusion:

The conclusion section will summarize the project's outcomes and provide an overall assessment of the success of the new restaurant management system. This section will cover the following topics:

- A summary of the project's objectives and the system's key features
- A summary of the testing results and evaluation of the system's performance
- A comparison of the new system with the client's previous system
- A discussion of any limitations or challenges encountered during the project
- Recommendations for future improvements or modifications to the system

Overall, the report will provide a detailed overview of the development and testing process for the new restaurant management system, as well as an assessment of its effectiveness and usability. The report will be structured in a logical and organized manner, with each section building on the previous one to provide a comprehensive understanding of the project and its outcomes.

2 Methodology

2.1 Requirements

By carrying out Requirements engineering;

2.1.1 Feasibility study

Through the study we found out that the business is a medium scaled one and not a large scaled one therefore, software addressing the exact requirements of medium scaled restaurants are very limited and even those plenty of software does not suit the budget.

The business also has a larger customer base to manage daily, and a potential growth can be observed.

Therefore, conclusion that we will be using MERN technology as it can be implemented using a common language, JavaScript which will be much easier for us to learn and use. MERN stack also allows us for rapid development with the use of modular design and pre-built components. Flexibility can be used to develop from simple applications to complex applications. This is also economically feasible as the entire stack uses open-source technologies, there is no need to purchase expensive licenses, making it a cost-effective option. MERN provides higher response time and cross-platform compatibility. Additionally, MERN has a large and active developer team. Overall MERN is a flexible technology which provides wide range of benefits to developers.

In conclusion, we identified that the best technology to be used to satisfy our client's requirements is the MERN stack.

2.1.2 Requirements Elicitation and Analysis

In the proposed system the stakeholders will be mainly, the owners, administrators, cashiers, and inventory manager from the phase of the management. On the other phase, customers, and drivers.

By conducting online meetings with the client, we identified their problems and the requirements as below:

- The existing system could not produce reports for analyzation daily, monthly, or yearly.
- Noticed inventory wastages and inventory fraud.
- Online customer platform for online ordering is not available.
- Delivery service not available
- Too complex for the staff members to use the software.

Through these requirements gathering, we identified the key features required for the system:

- Inventory control system.
- Online platform for the customers to place their orders.
- Delivery system.
- Order management.
- User management.
- Frequently asked questions.
- Waiter management.
- Reporting.

2.1.3 **Requirement Specification**

- *Functional requirements*
 1. The system should allow customers to browse the menu, place orders, and make reservations online.
 2. The system should provide restaurant staff with real-time updates on order status, table assignments, and customer requests,
 3. The system should provide automatic notifications for staff when orders are ready.
 4. The system should allow staff to manage inventory levels and receive alerts when supplies need to be replenished.
 5. The system should provide reporting features, including sales reports, inventory reports, and customer satisfaction reports.
 6. The system should provide the ability to generate invoices.
- *Non-Functional Requirements*
 7. The system should be highly reliable, with minimal downtime or errors.
 8. The system should be scalable and able to handle large number of users and activities.
 9. The system should be secure, with user authentication and data encryption features.
 10. The system should be responsive and provide quick load times for web pages and orders.
 11. The system should be easy to maintain, with a clear and well-documented codebase.

By identifying and documenting the functional and non-functional requirements of the system, we then came up with the use case diagram.

Use case Diagram.

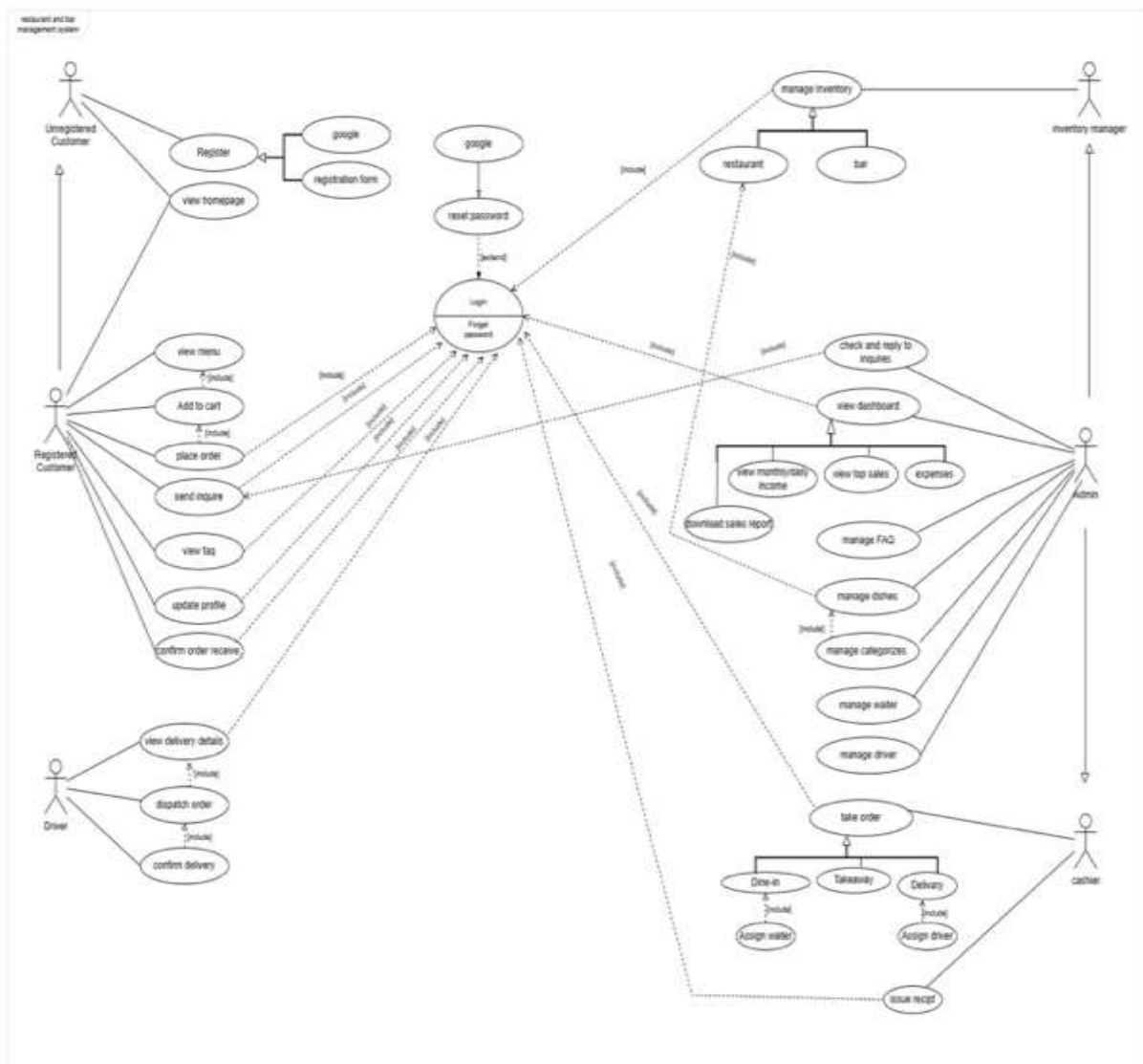


Figure 2.1 Use case diagram of the Restaurant Management System

2.1.4 Requirements validation

In this stage, we as a team verified that the software requirements are complete, accurate, and consistent with the needs and expectations of the stakeholders by reviewing requirements, identifying inconsistencies, prioritizing requirements, ensuring that each requirement can be tested individually and made note of any changes that were identified during the validation process.

This stage helped us to ensure that the system will be developed and delivered successfully.

2.2 Design

2.2.1 System Overview

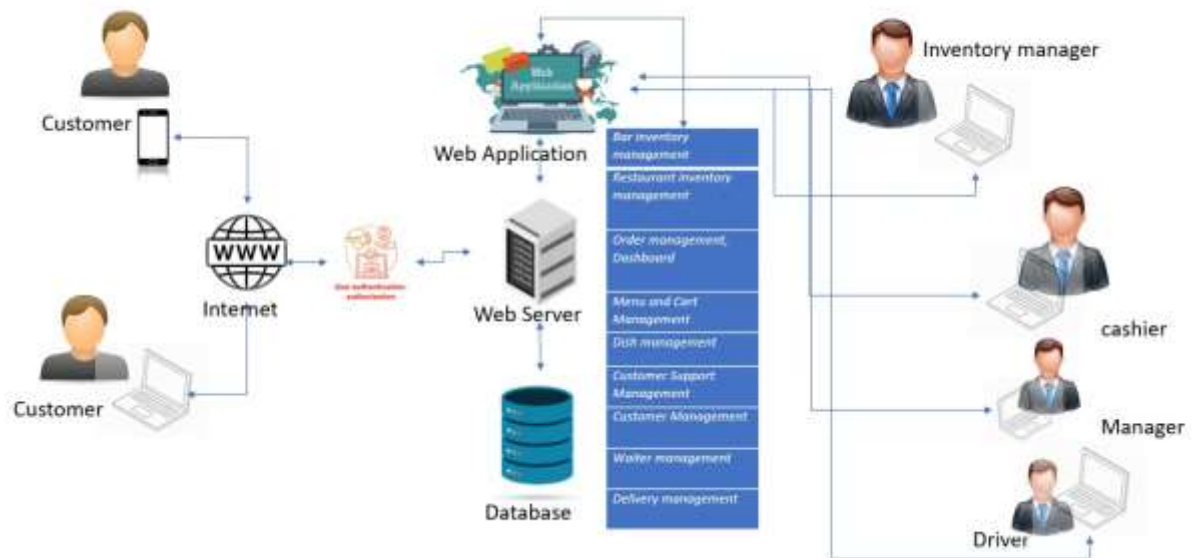


Figure 2.2 System Overview of the Restaurant management system

2.2.2 High-level architecture design

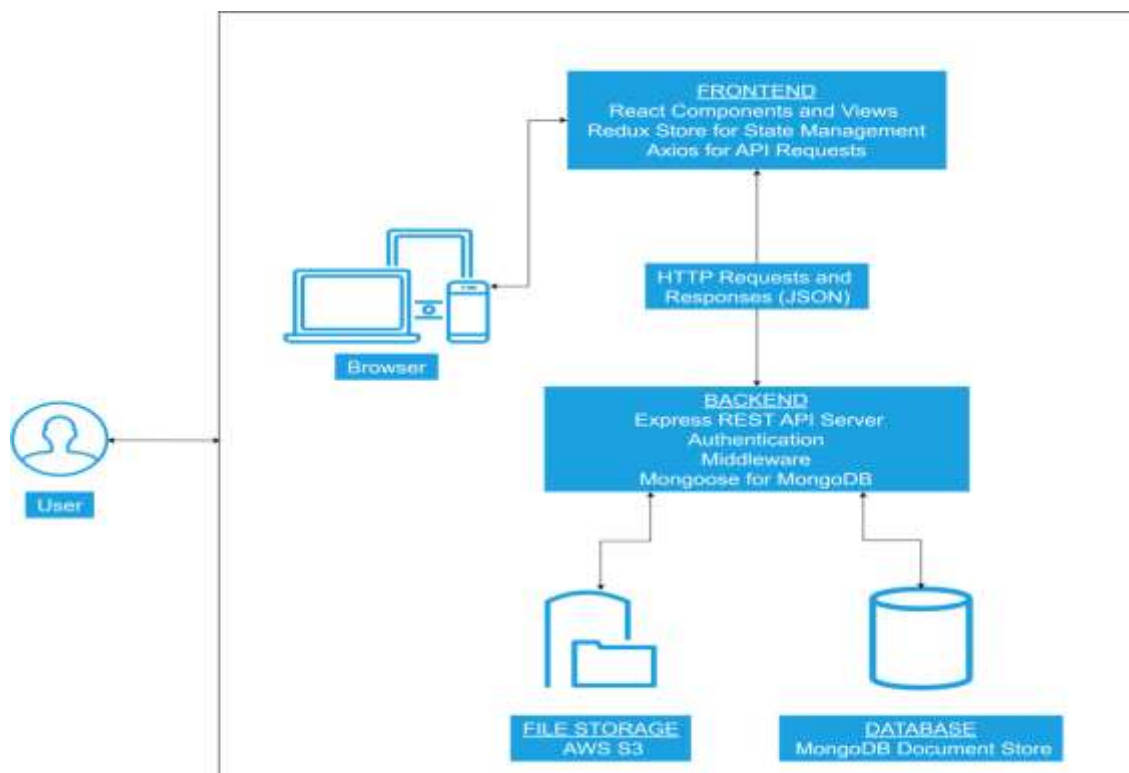


Figure 2.3 High-level architecture design

2.2.3 Network design

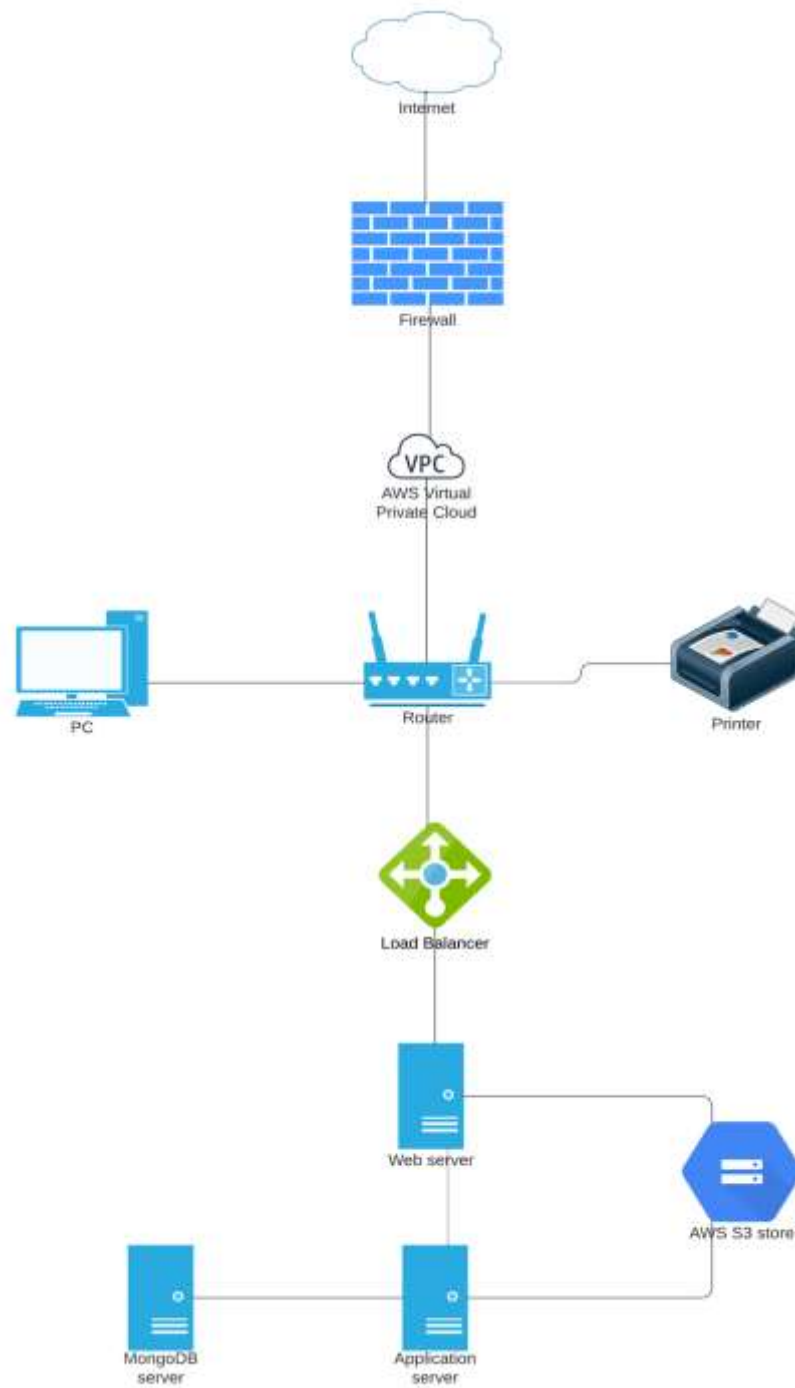


Figure 2.4 Network design

2.2.4 Class diagram

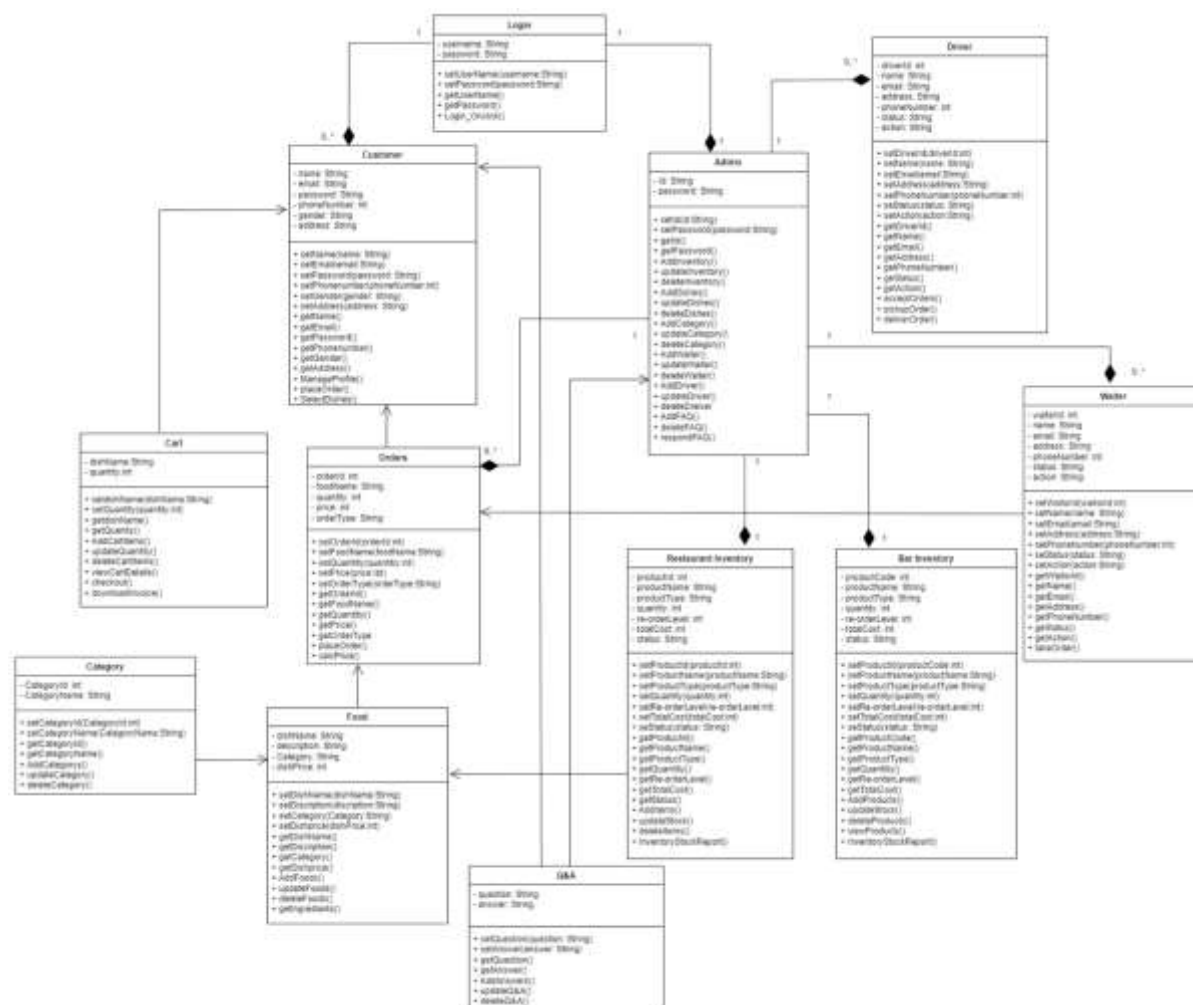


Figure 2.5 Class Diagram for Restaurant management system

2.2.5 Sequence Diagrams

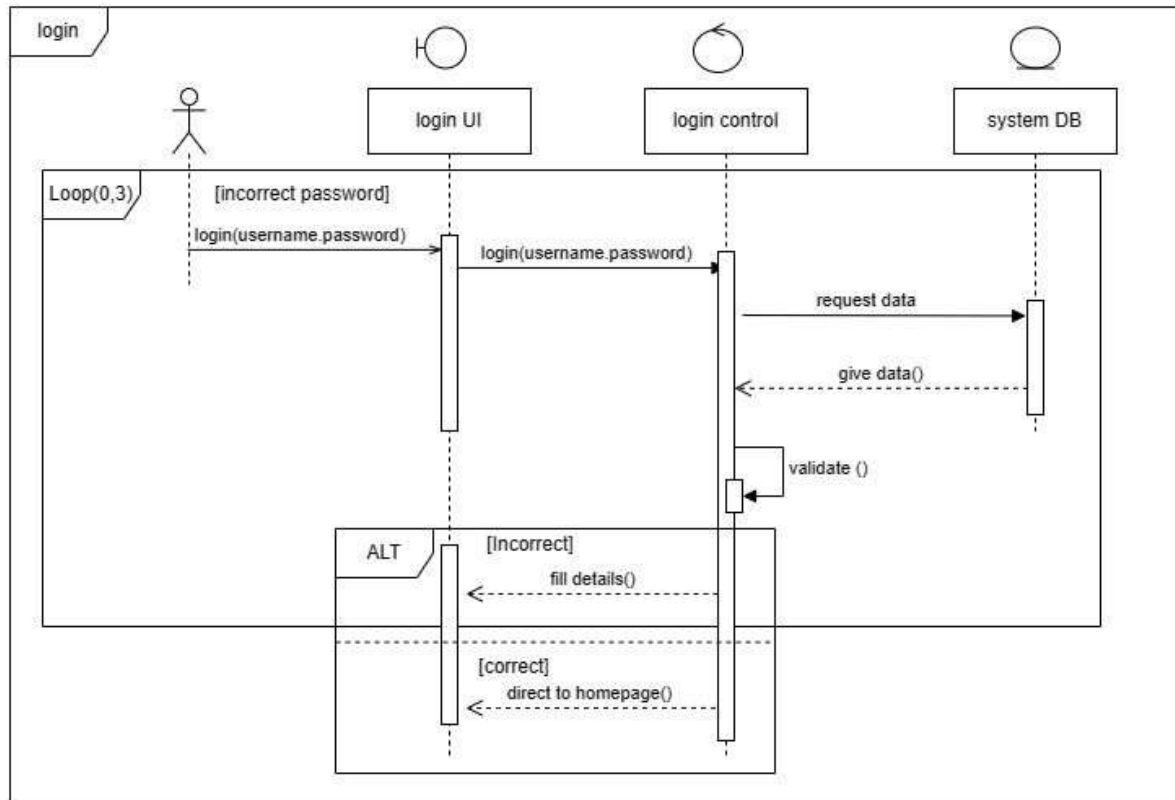


Figure 2.6 Sequence diagram for Customer login

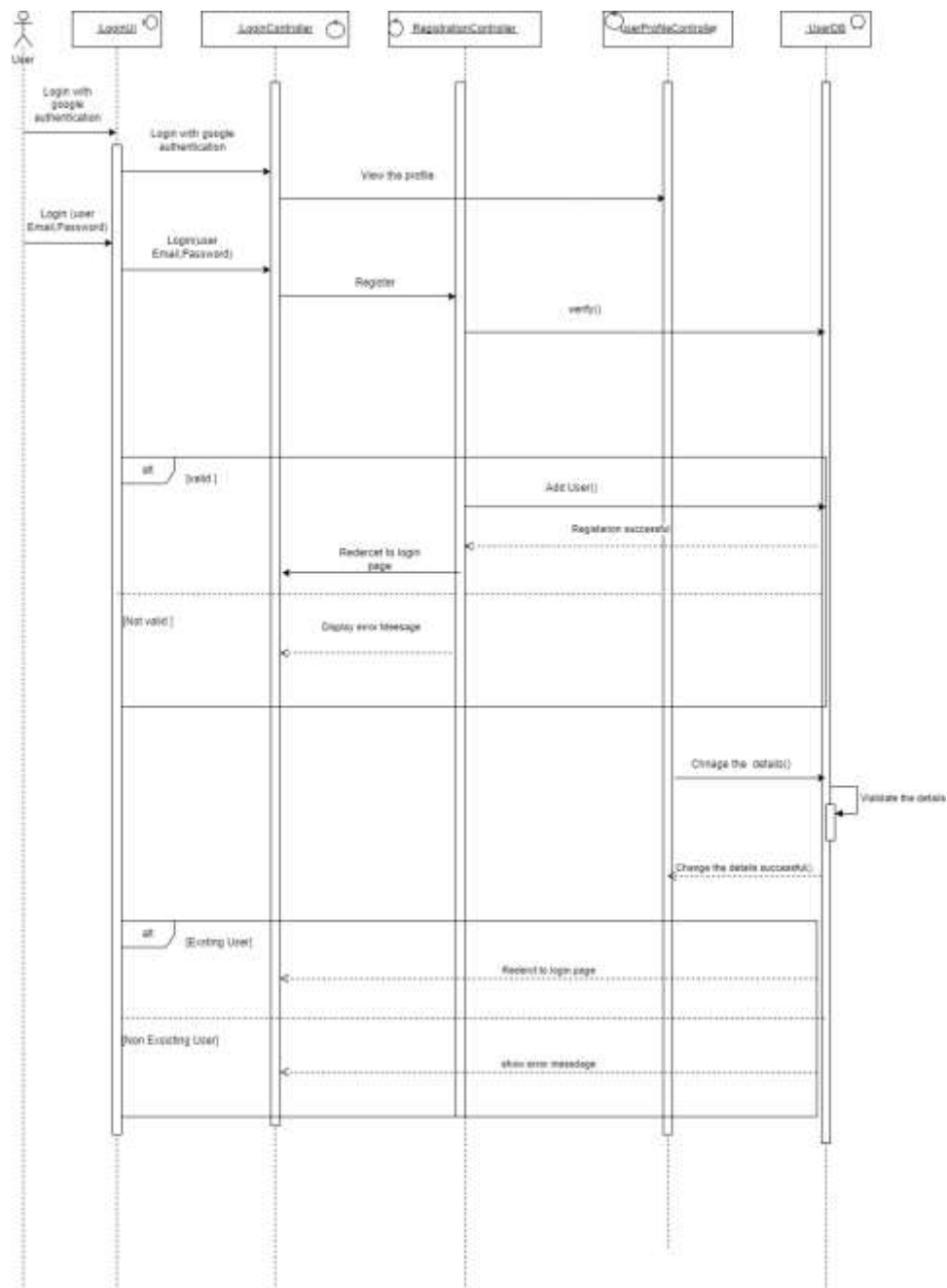


Figure 2.7 Sequence diagram for User management

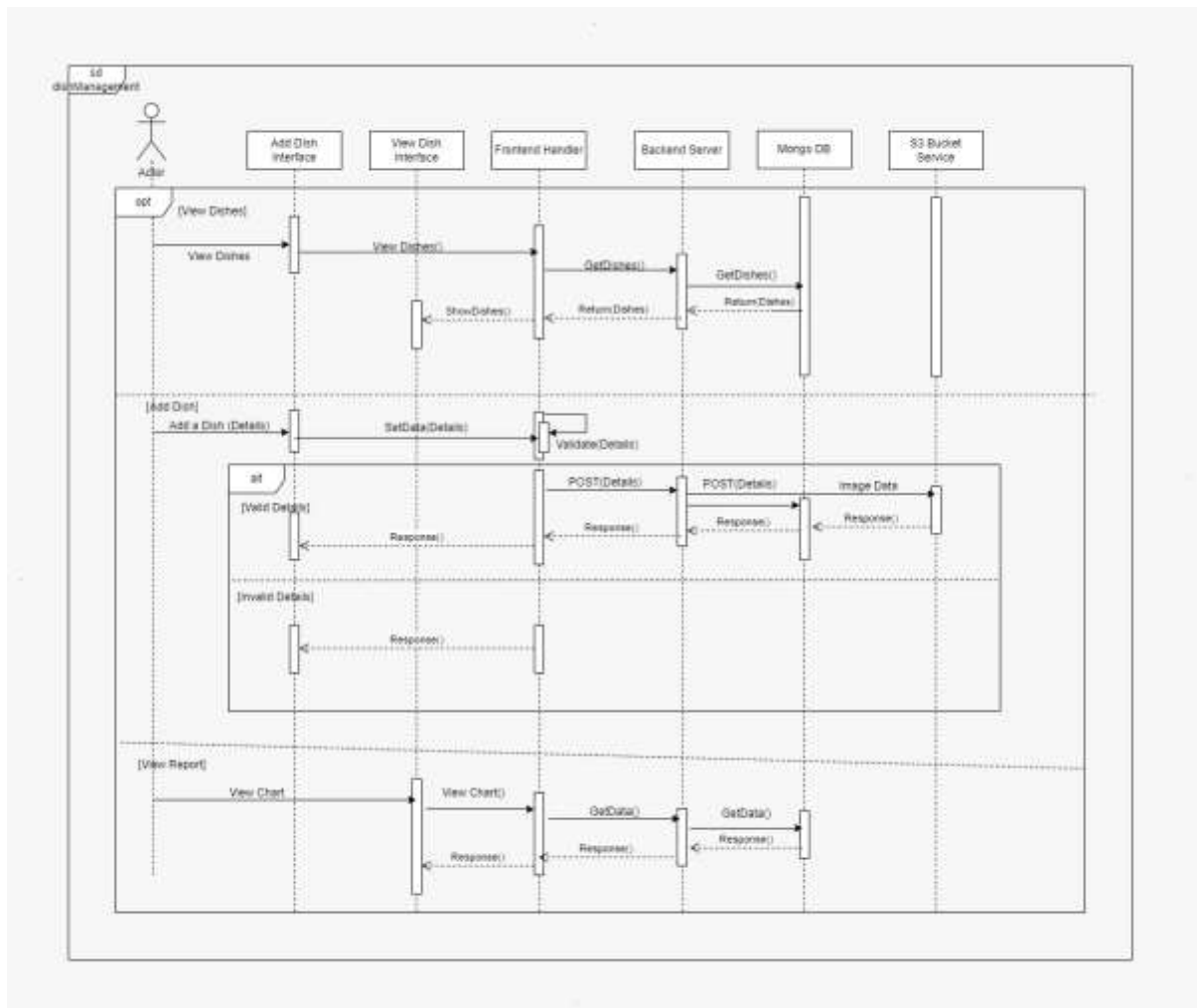


Figure 2.8 Sequence diagram for Dish management

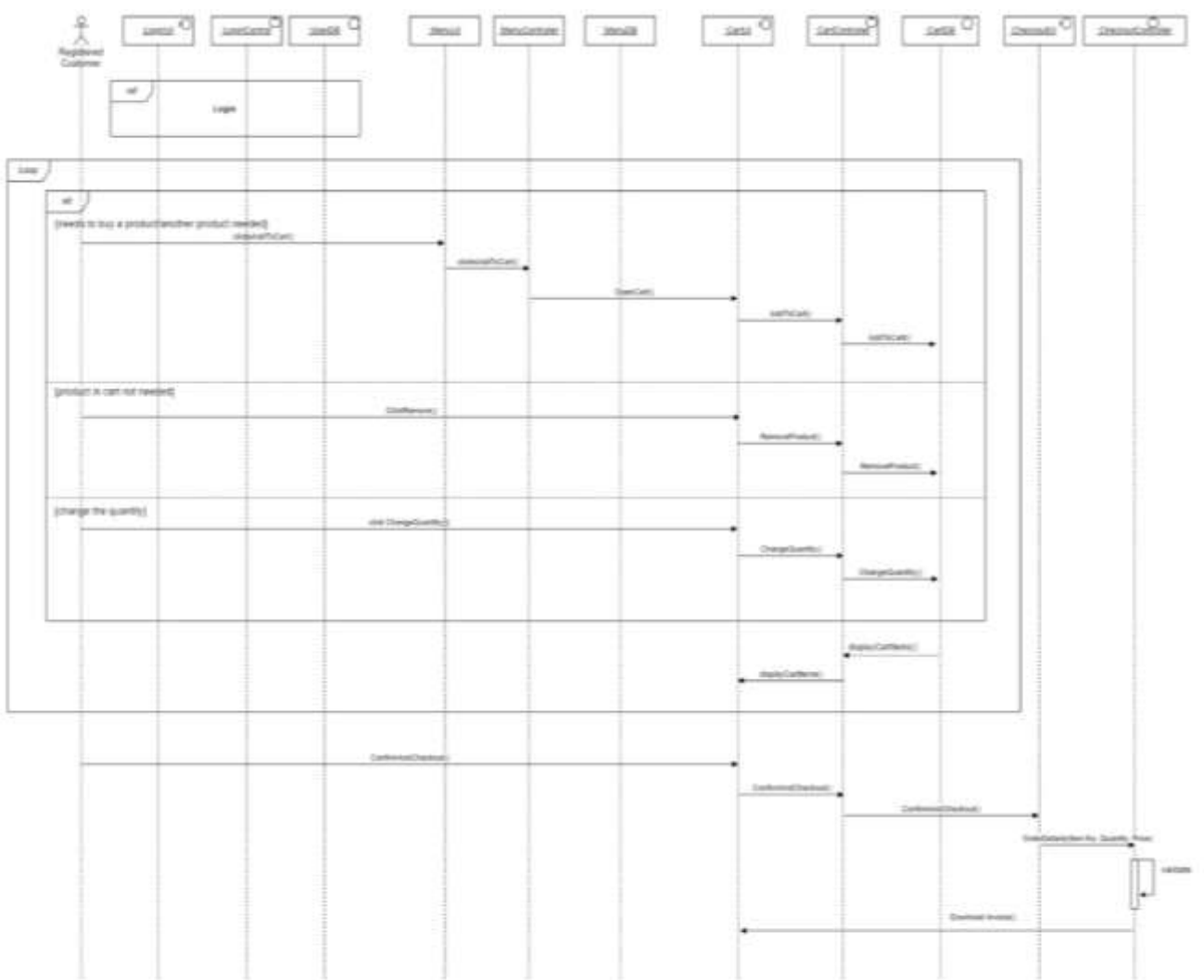


Figure 2.9 Sequence diagram of Menu and Cart management

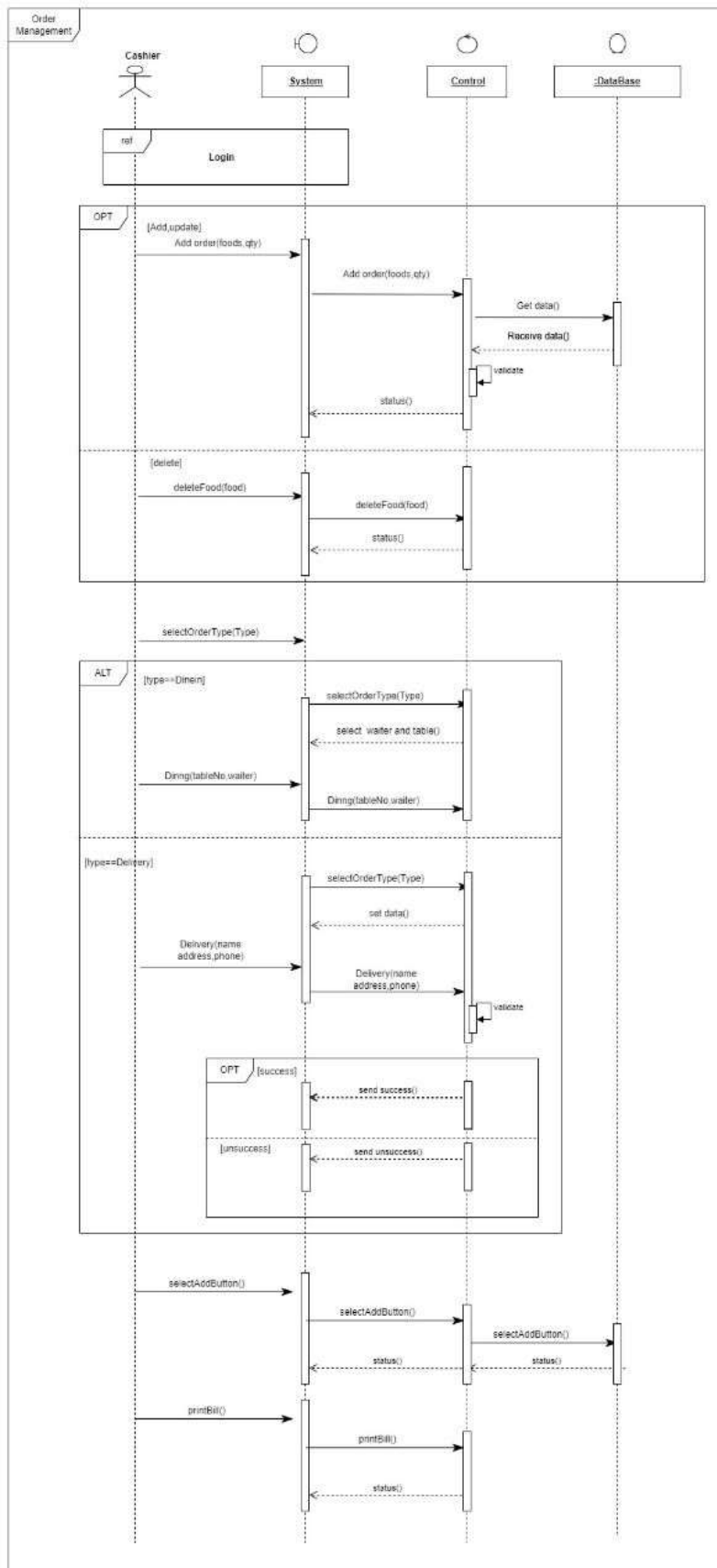


Figure 2.10 Sequence diagram for placing an order

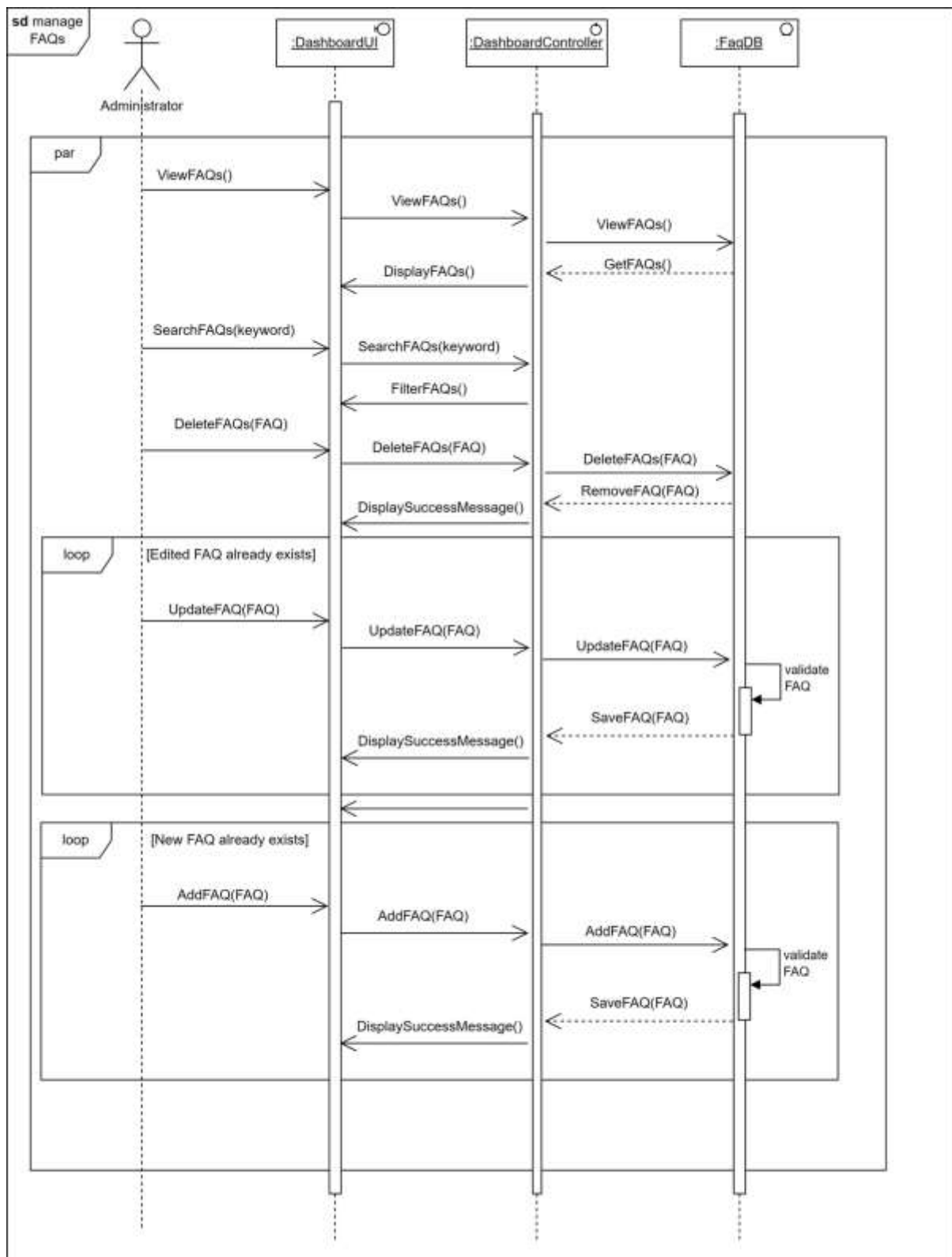


Figure 2.11 Sequence diagram for FAQ management

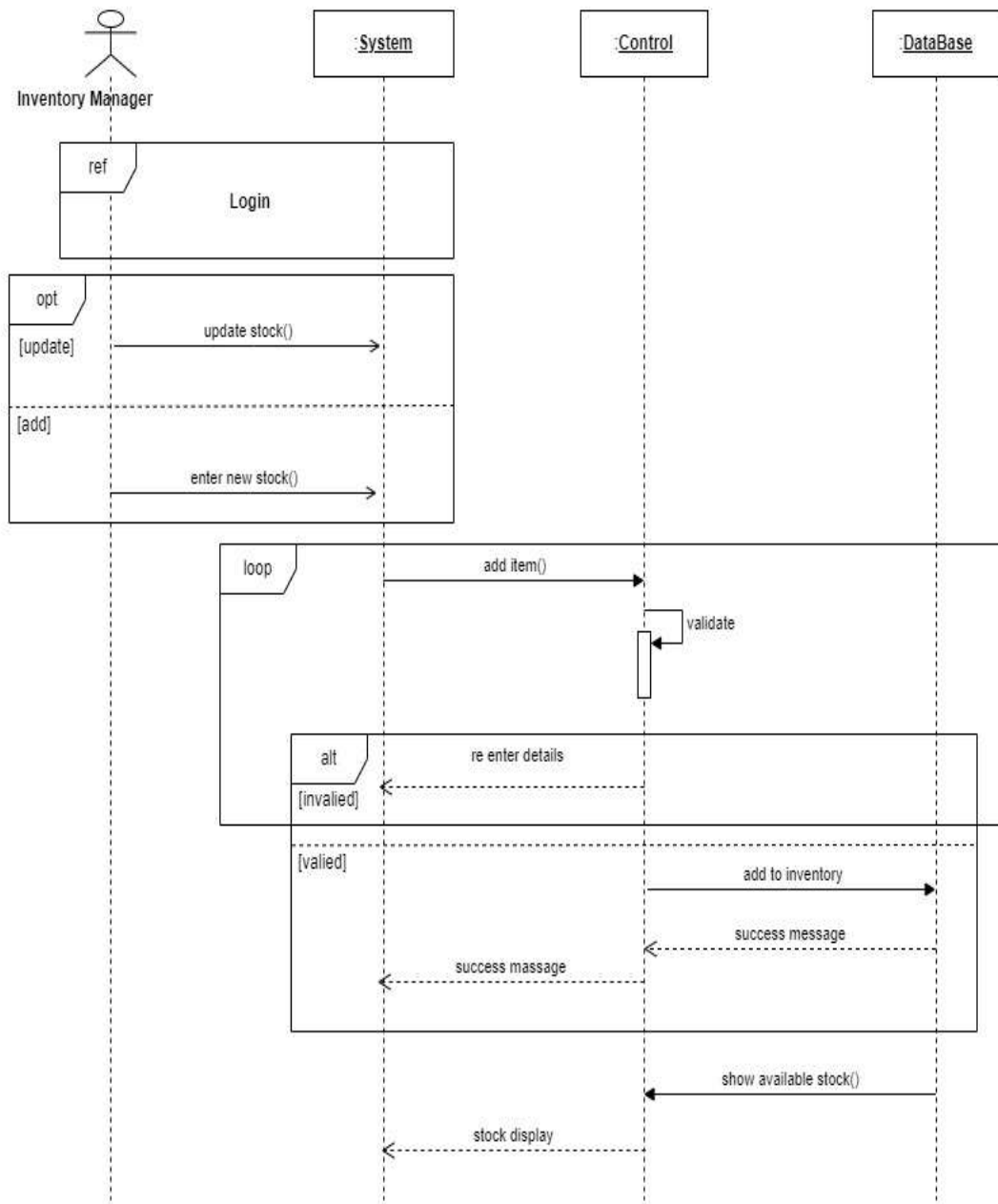


Figure 2.12 Sequence diagram for Restaurant Inventory management

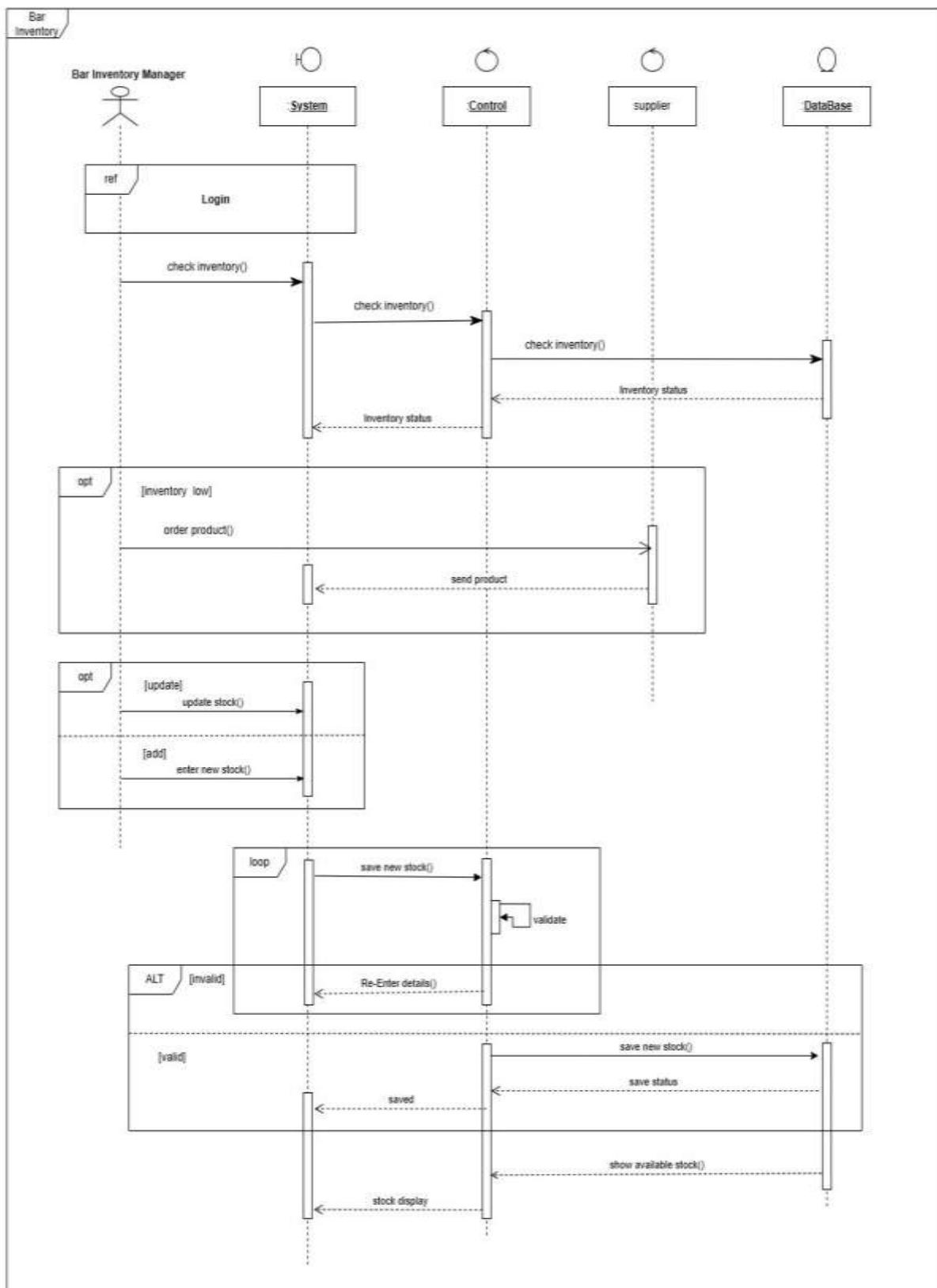


Figure 2.13 Sequence diagram for Bar Inventory management

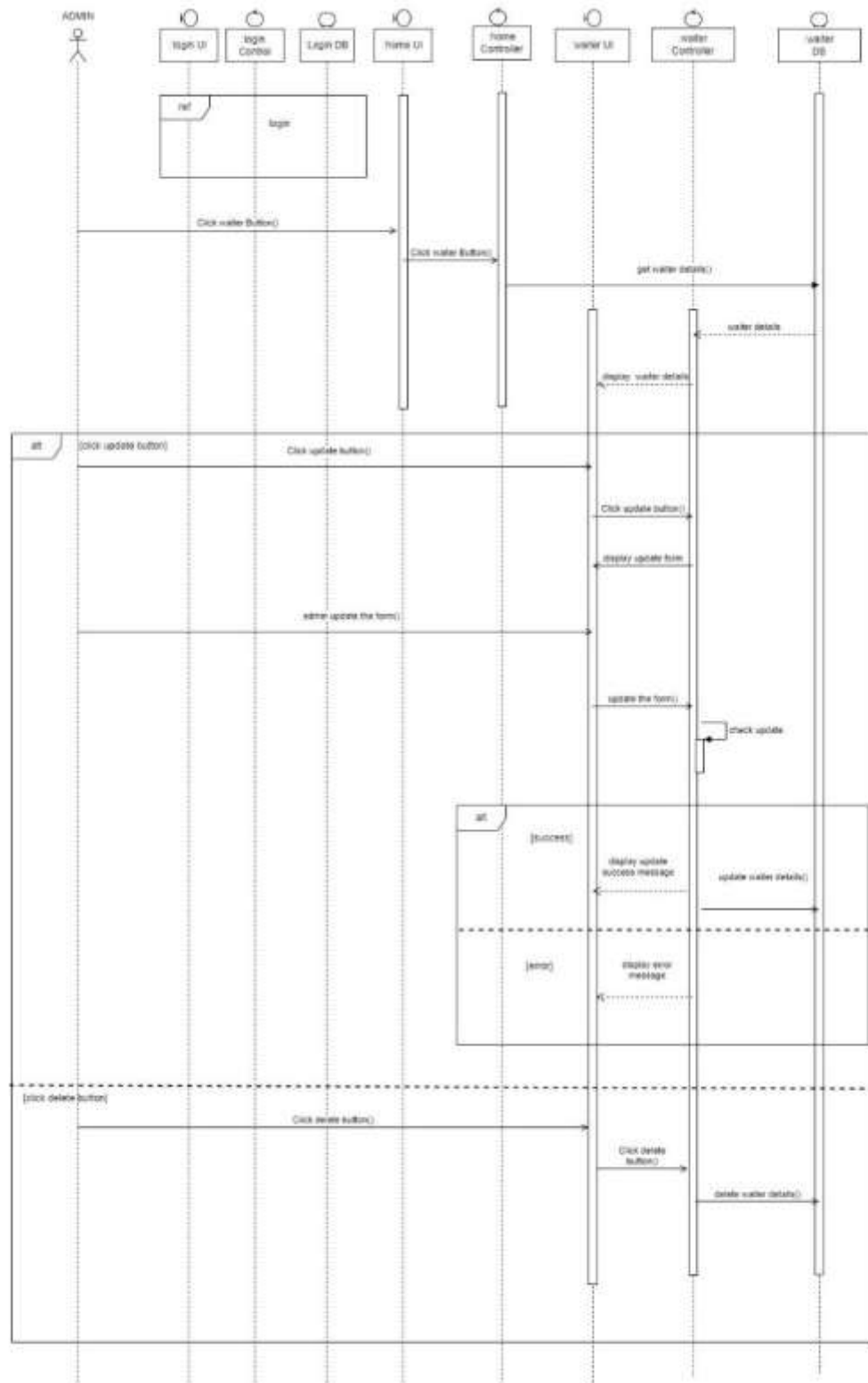


Figure 2.14 Sequence diagram for Waiter management

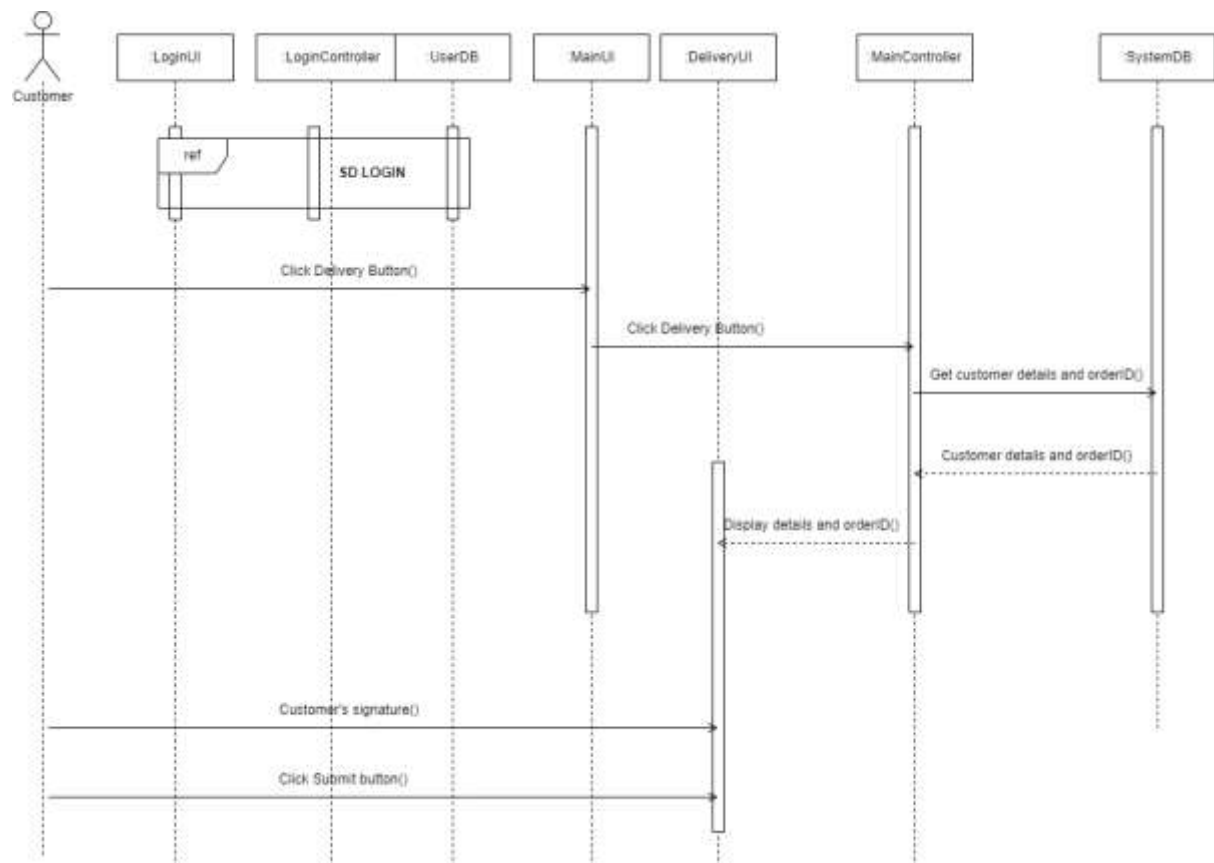


Figure 2.15 Sequence diagram for Delivery management

2.2.6 Activity Diagrams

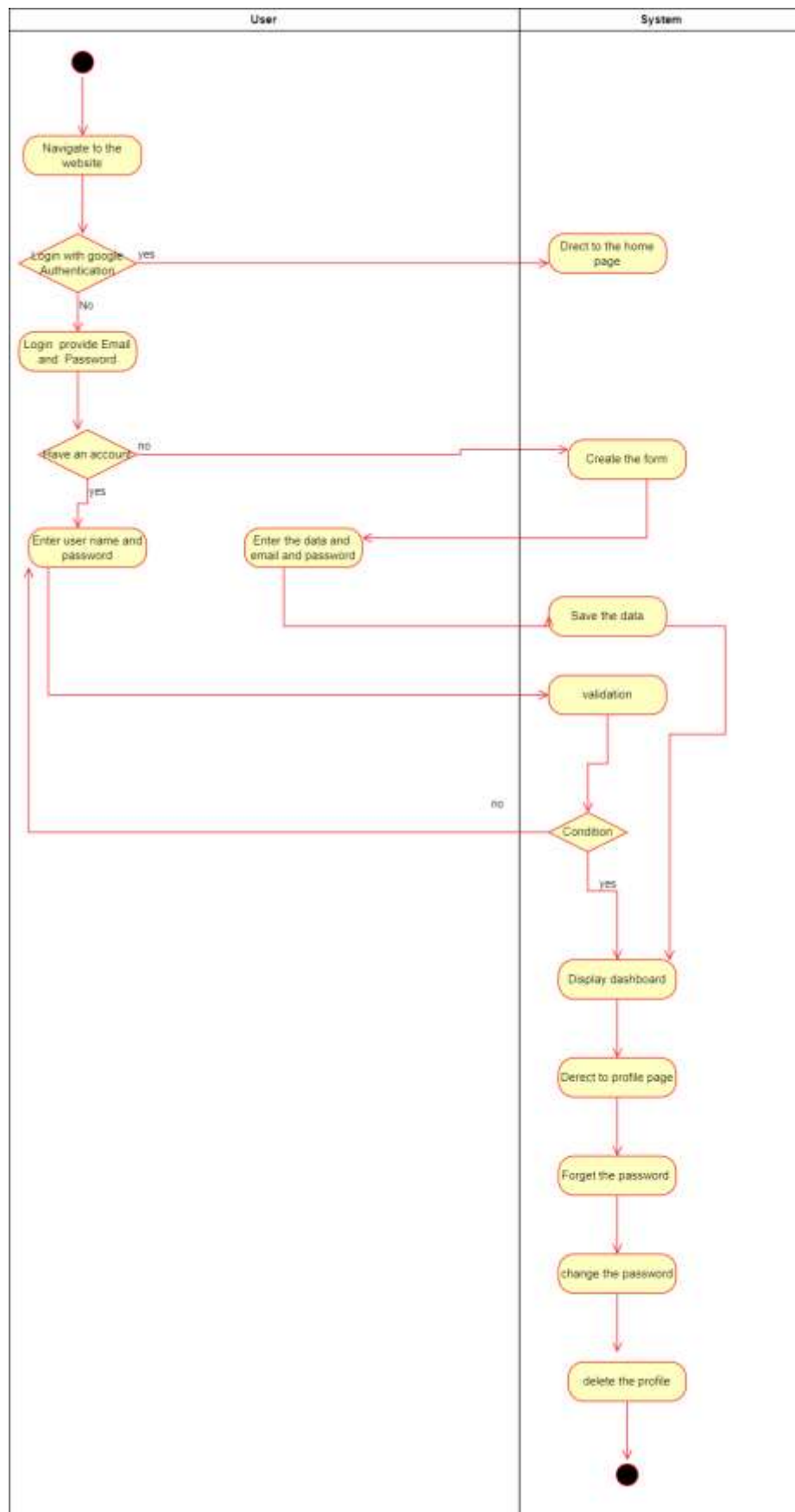


Figure 2.16 Activity diagram for Customer Login

Figure 2.17
Activity diagram
Dish
management

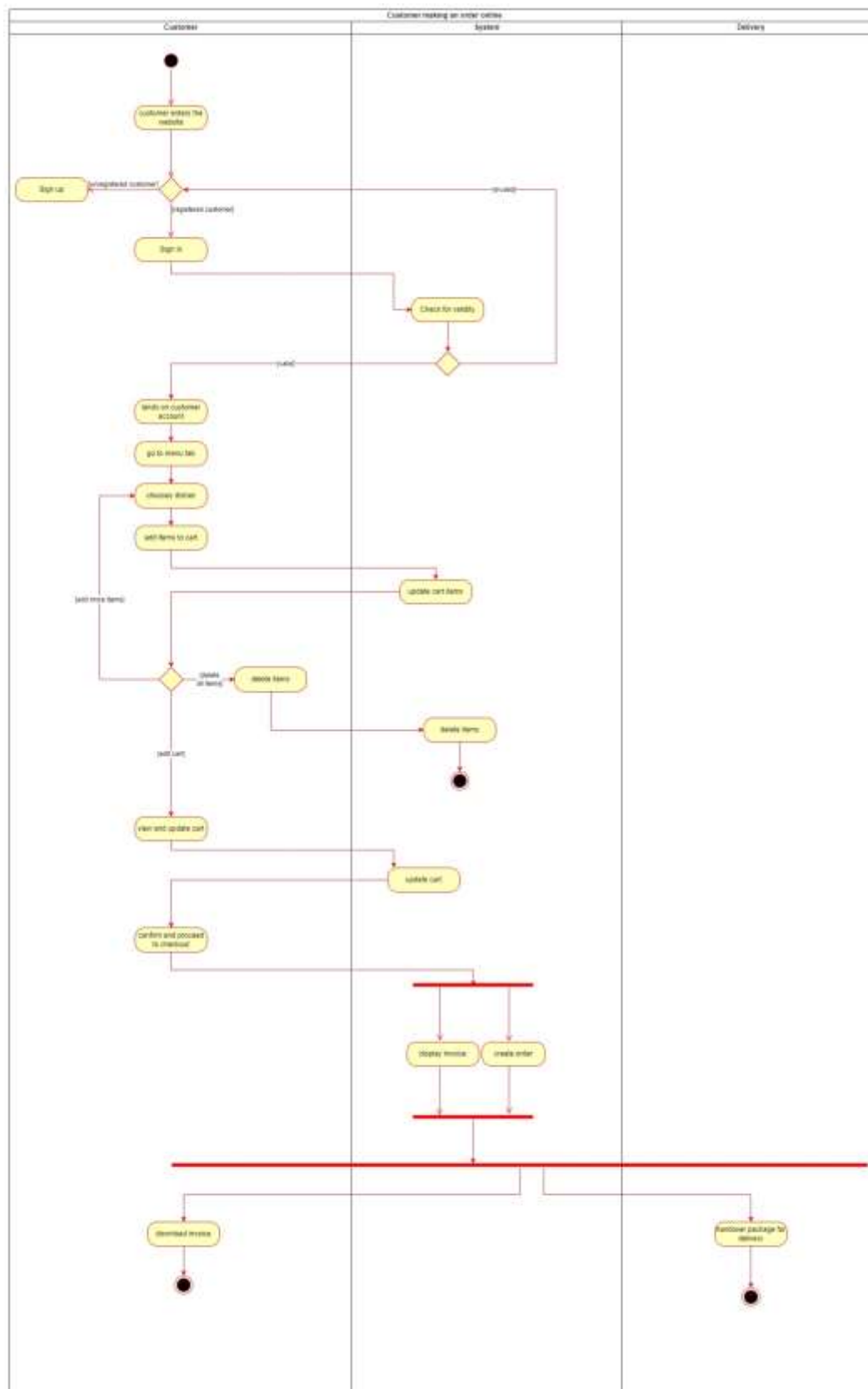


Figure 2.18 Activity diagram for Customer placing an order online

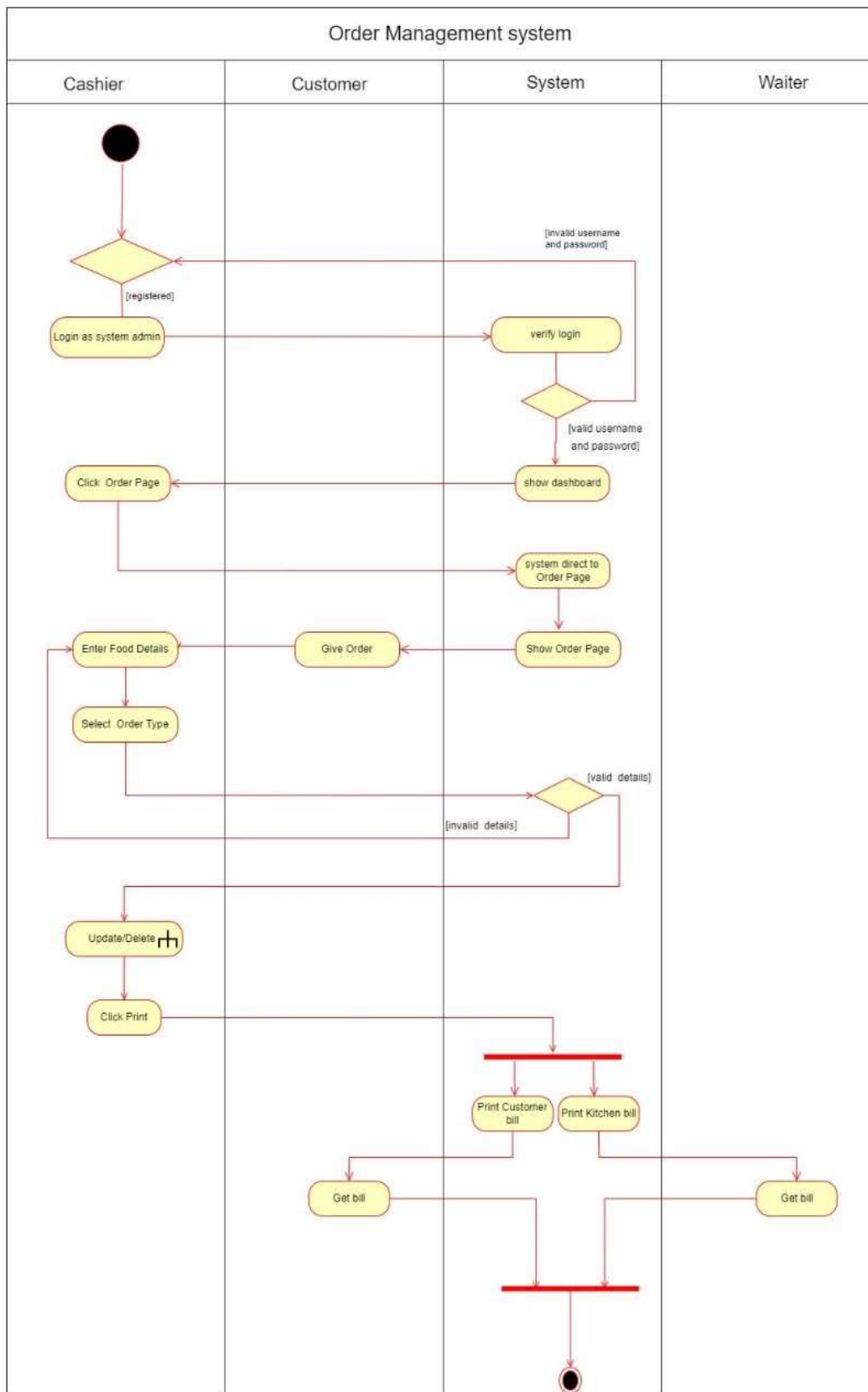


Figure 2.19 Activity diagram for placing an order

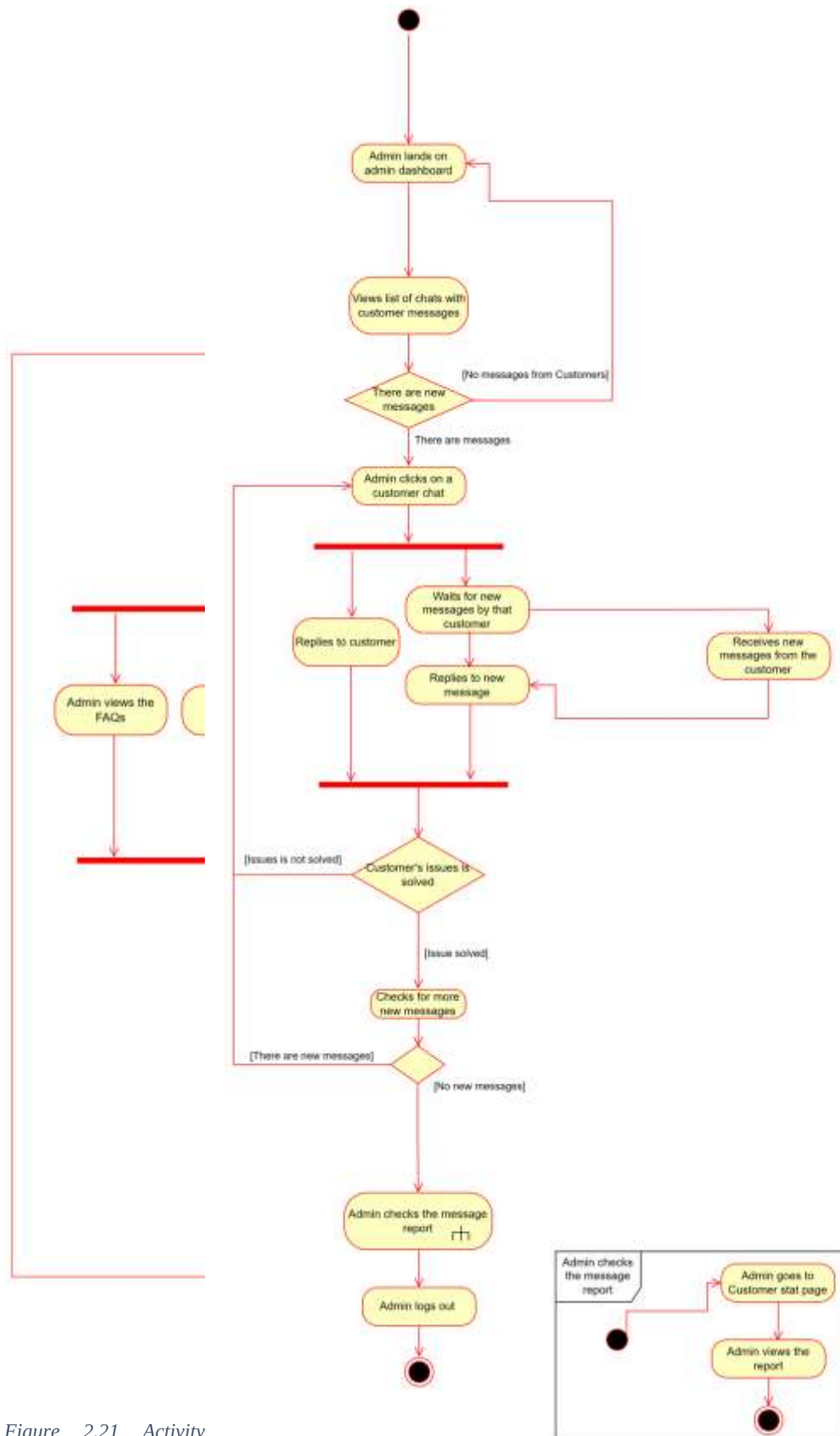


Figure 2.21 Activity diagram of managing FAQs

Figure 2.20 Activity diagram of Admin replying to direct messages

Admin

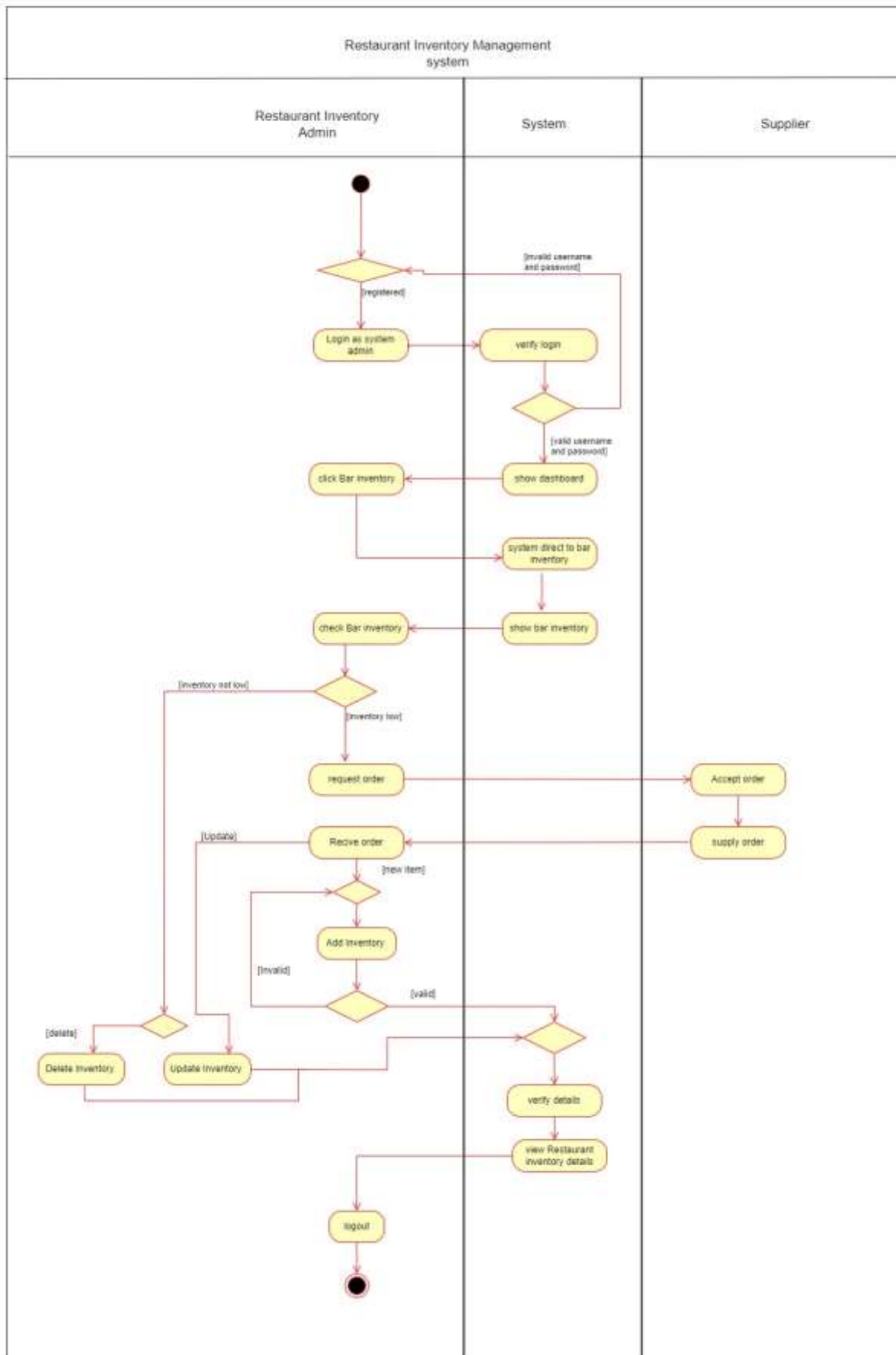


Figure 2.22 Activity diagram for Restaurant Inventory management

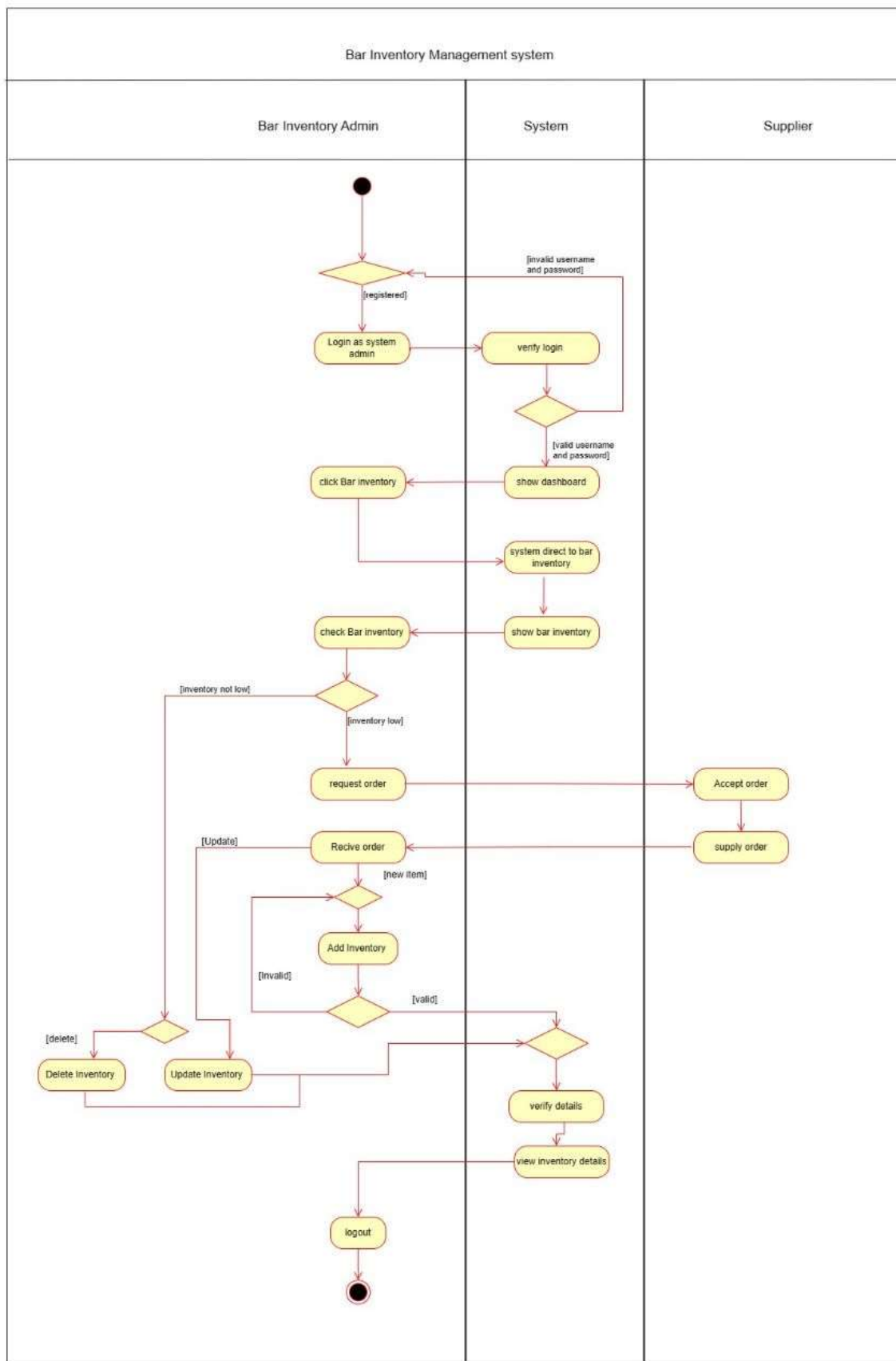


Figure 2.23 Activity diagram for Bar inventory management

2.2.7 ER Diagram

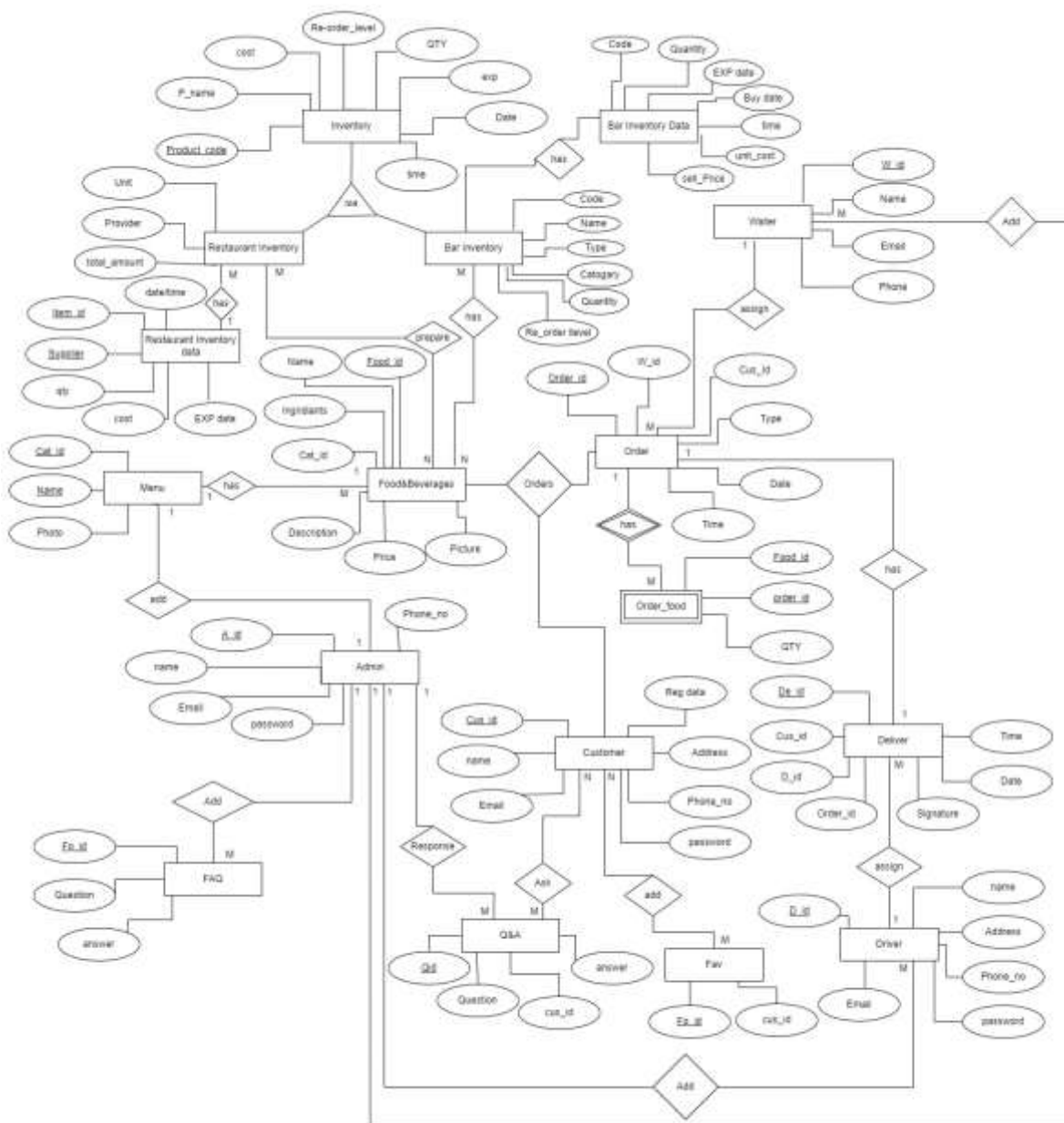


Figure 2.24 ER diagram for Restaurant management system

2.2.8 Normalized Schema



2.2.9 User interfaces

Customer Phase

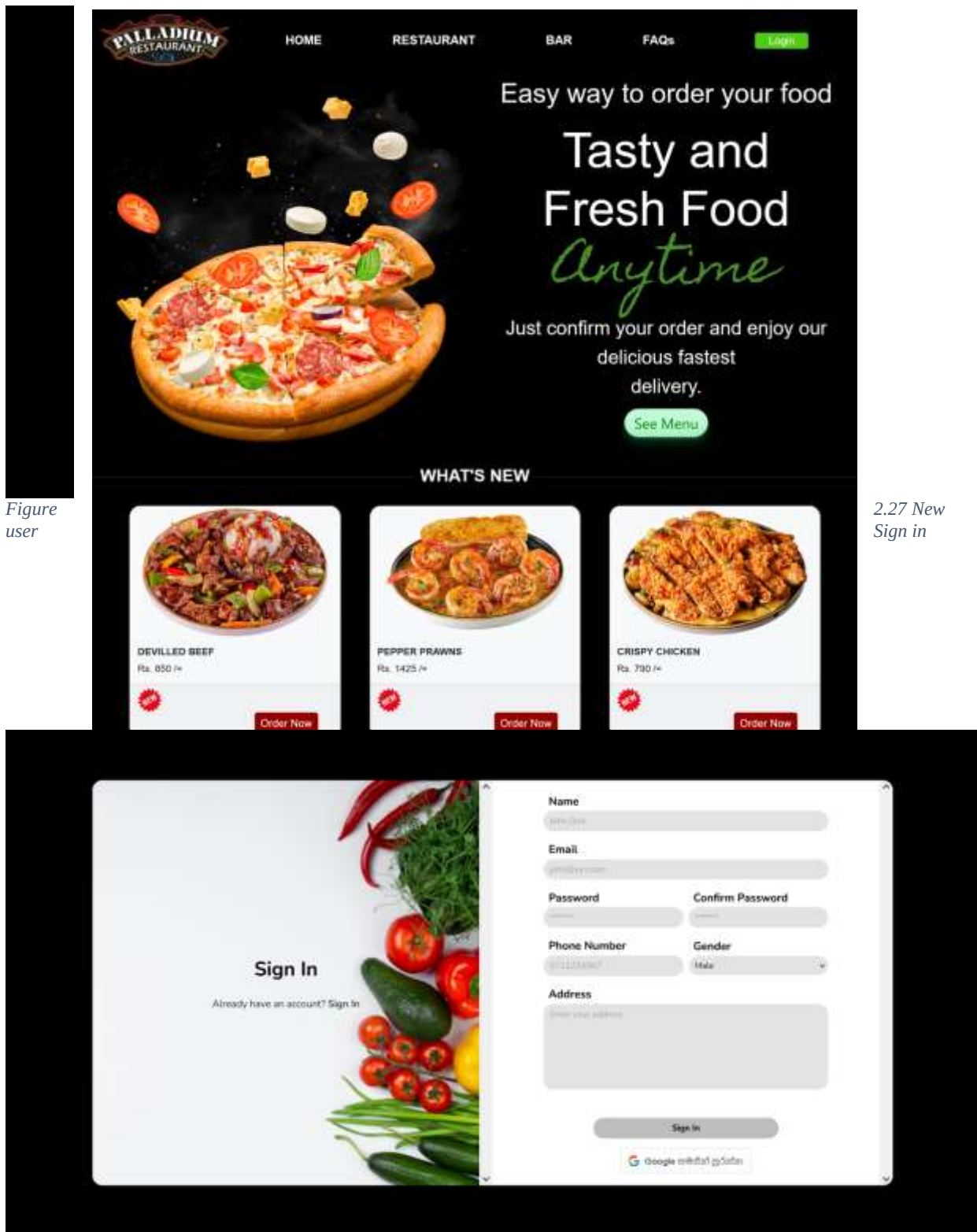
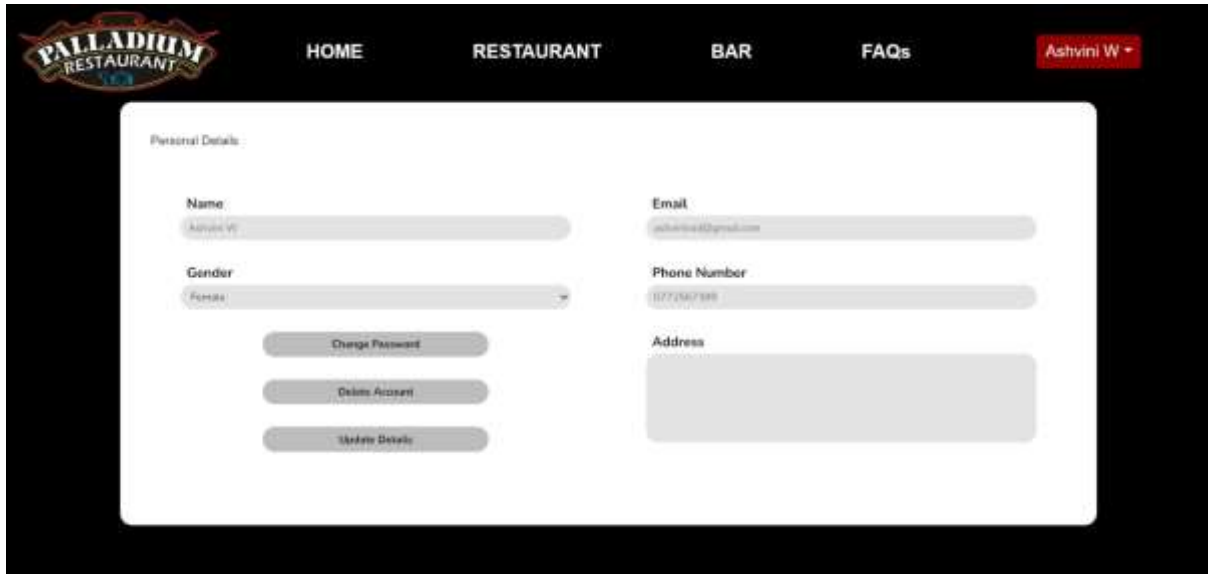


Figure 2.26 Registered user Sign in

Figure 2.25 Homepage



The screenshot shows a web application for "PALLADIUM RESTAURANT". The navigation bar includes links for HOME, RESTAURANT, BAR, and FAQs. A user profile dropdown menu is open, showing the name "Ashvini W" with a plus icon. The main content area is titled "Personal Details" and contains a form with the following fields and buttons:

- Name:** A text input field containing "Ashvini W".
- Email:** A text input field containing "ashviniw@gmail.com".
- Gender:** A dropdown menu with "Female" selected.
- Phone Number:** A text input field containing "077/567585".
- Address:** A large text input field.
- Buttons:** Three buttons are located below the Name and Gender fields: "Change Password", "Delete Account", and "Update Details".

Figure 2.28 User profile management

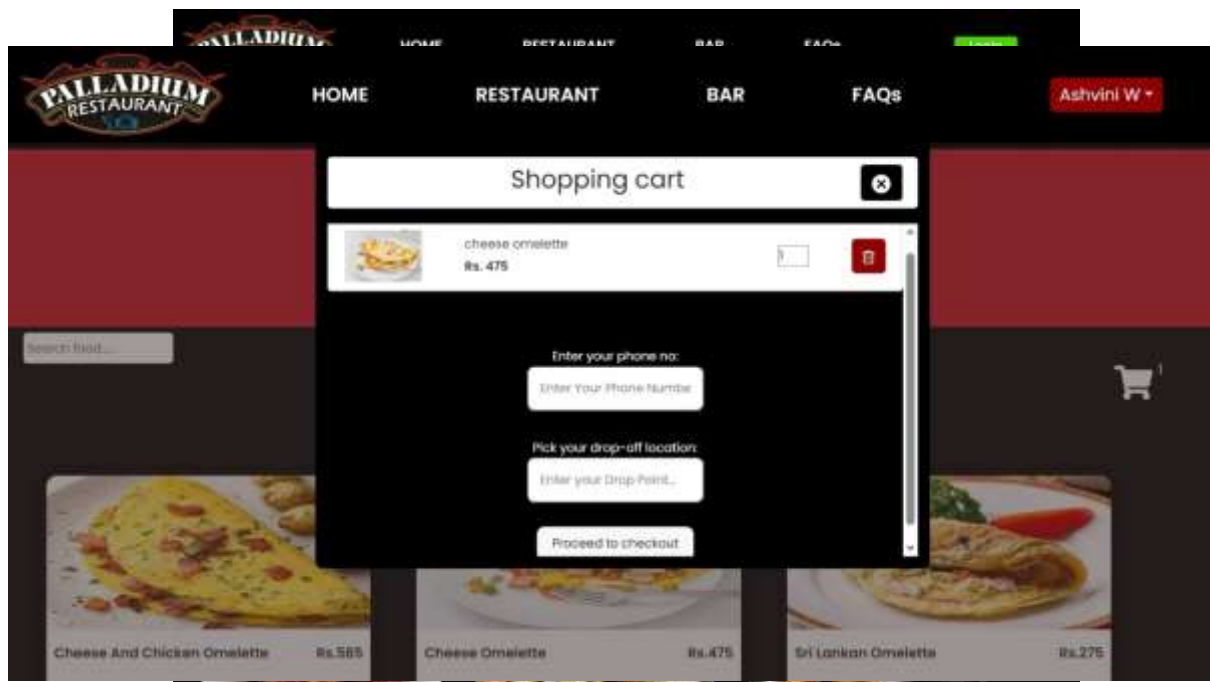


Figure 2.30 Shopping cart

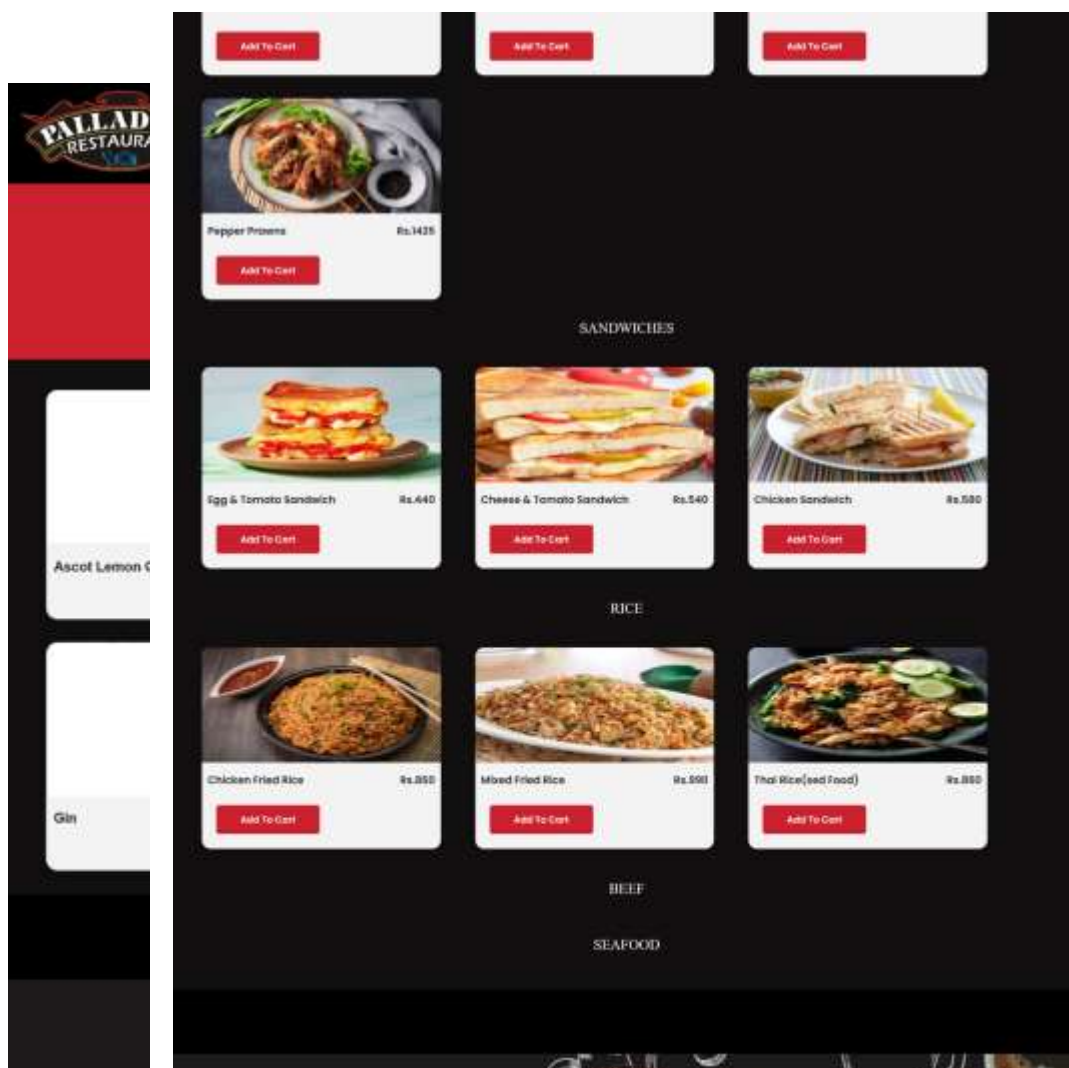


Figure 2.31

Figure 2.29 Restaurant Menu sorted into Categories

Bar item

display

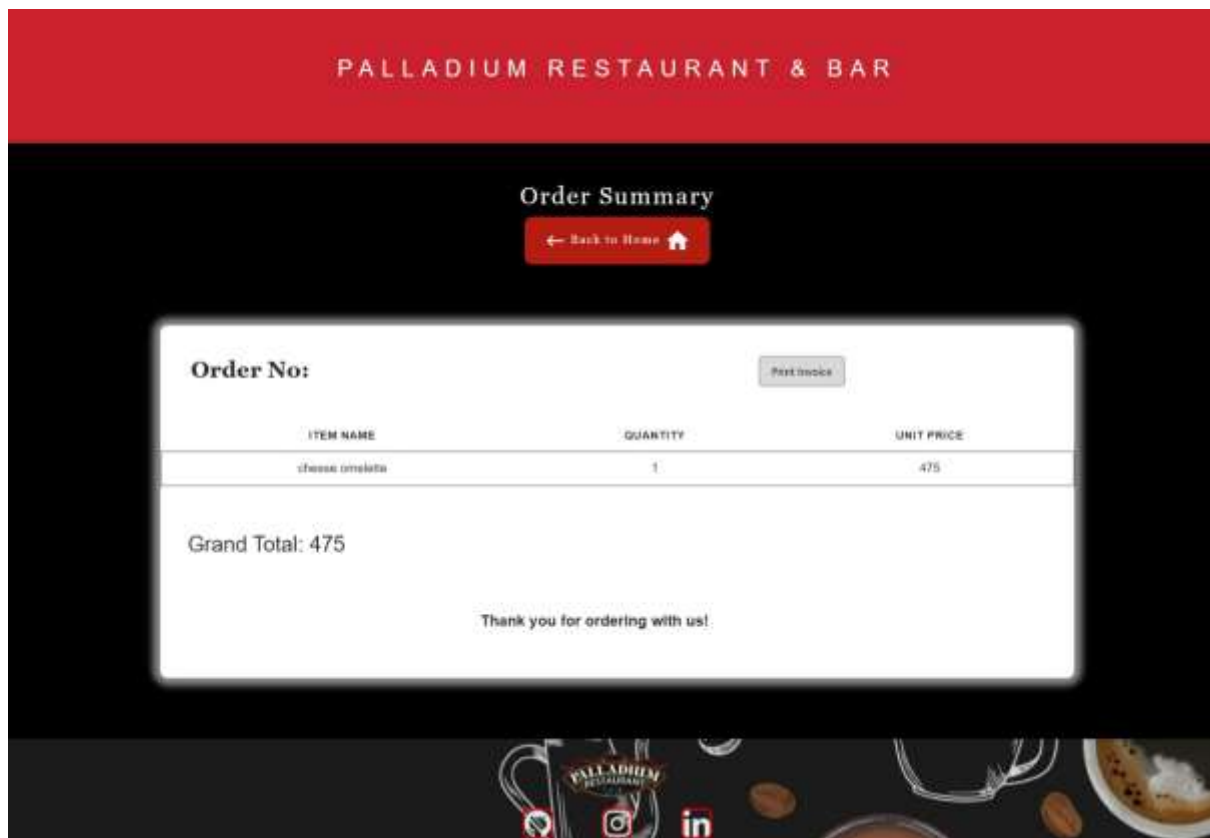


Figure 2.32 Order Summary

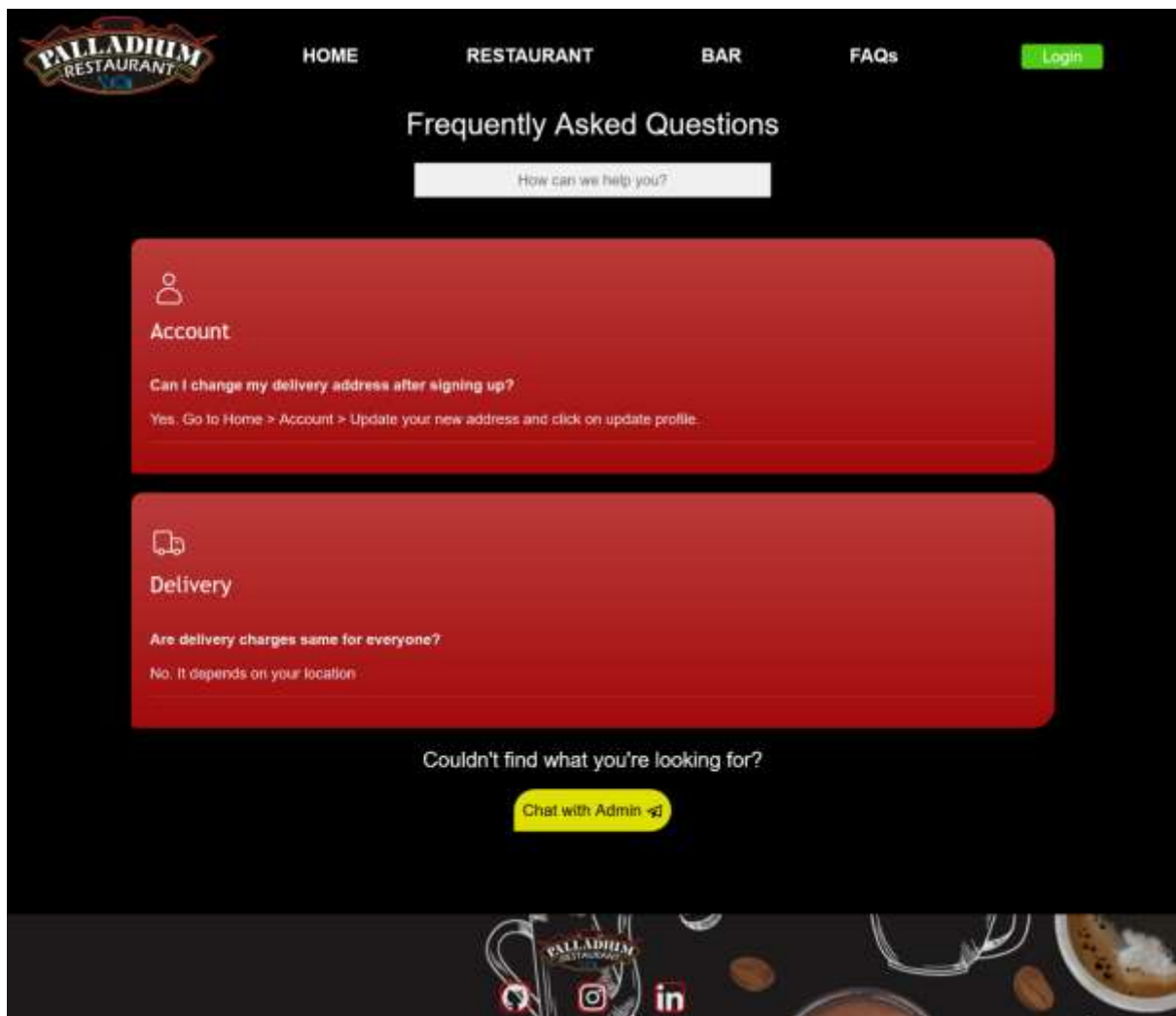


Figure 2.33 FAQ display for Customers

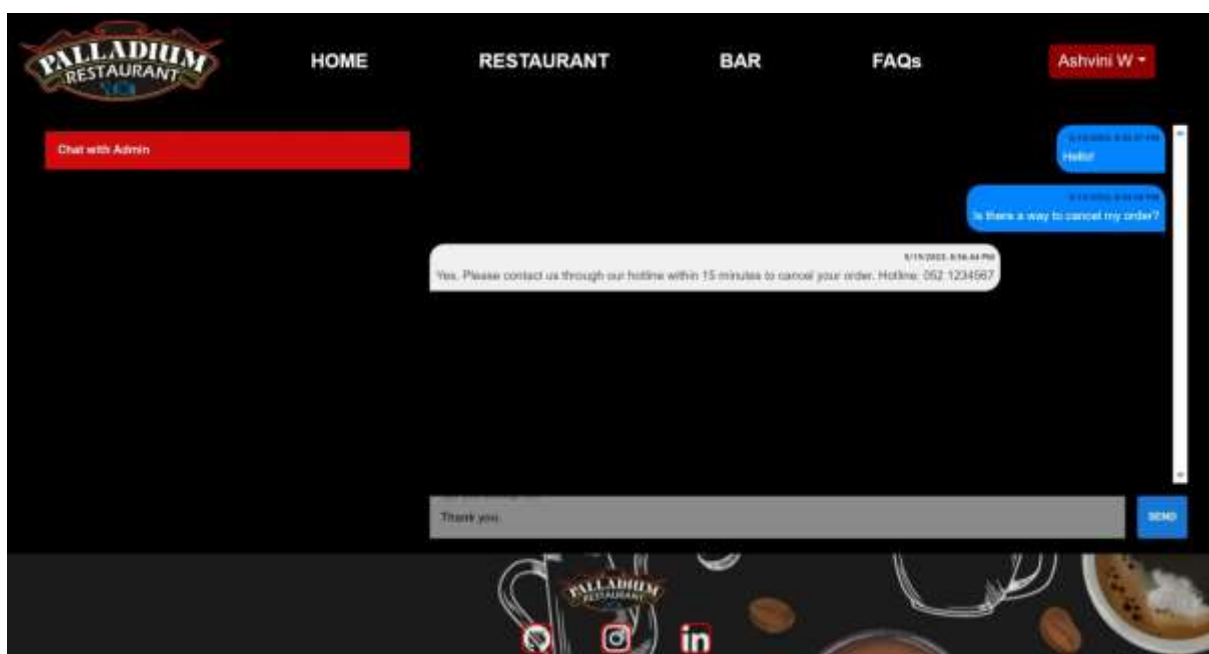


Figure 2.34 Customer live chat

Admin Phase

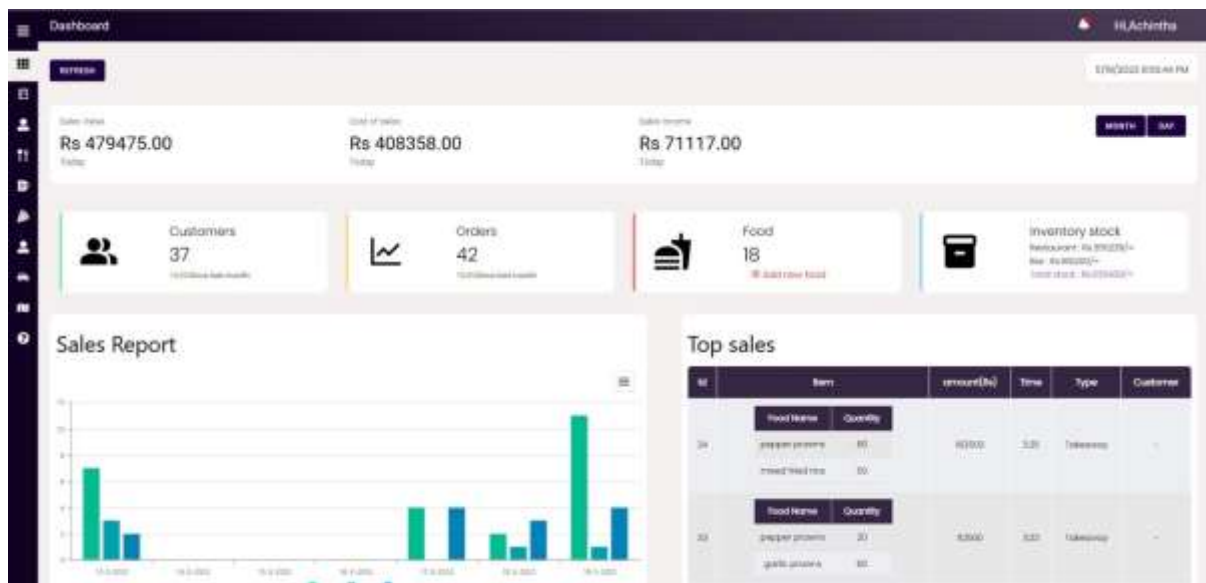


Figure 2.36 Admin Dashboard

The Dish Management interface allows administrators to add new dishes to the menu. It includes a sidebar with navigation options and a main content area with a form for adding a dish.

Add a Dish Form:

- Dish Name:** devilled prawns
- Category:** Prawns
- Price:** 990

Select the ingredients for dish:

Ingredient ID	Ingredient Name	Quantity Left	Cost for ingredient	Set unit	Set Quantity	Action
11	onion	~808	8940	g	120	Add

Buttons: Go Back, View Dishes, Select a picture of the Dish (Choose File: devilled prawns.jpg), + Add Dish

Added Ingredients will show here:

Name	Quantity	Unit	Price	Remove
prawns	360	g	-20794863006208483	Remove

Total Cost for the Dish's Ingredients: -305.878252308386

Figure 2.35 Dish Management

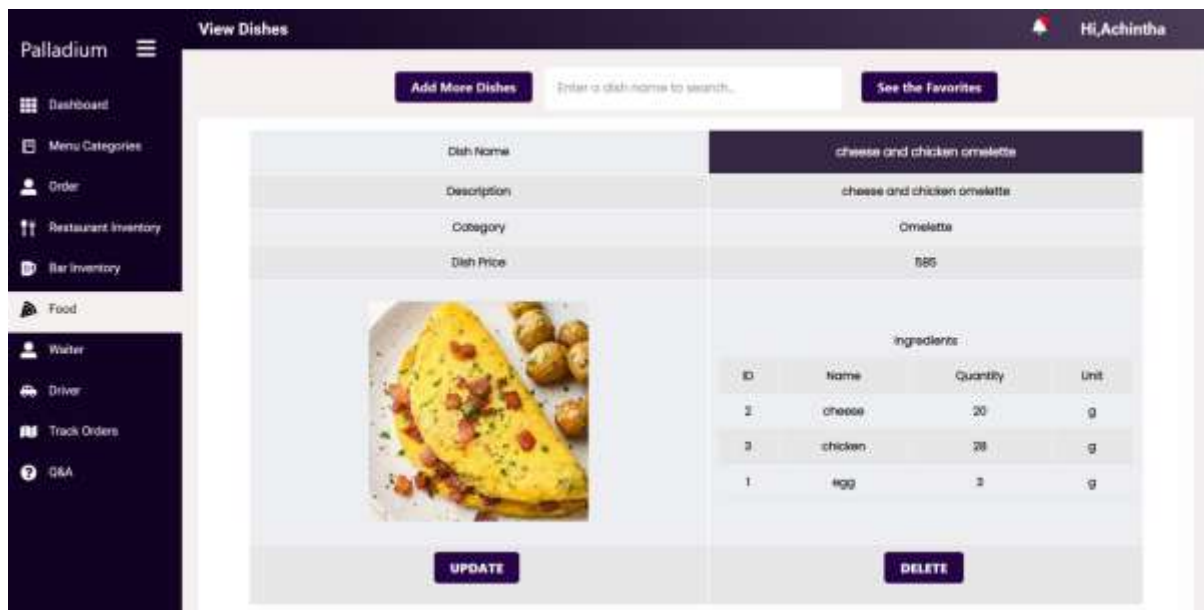


Figure 2.37 Dish management

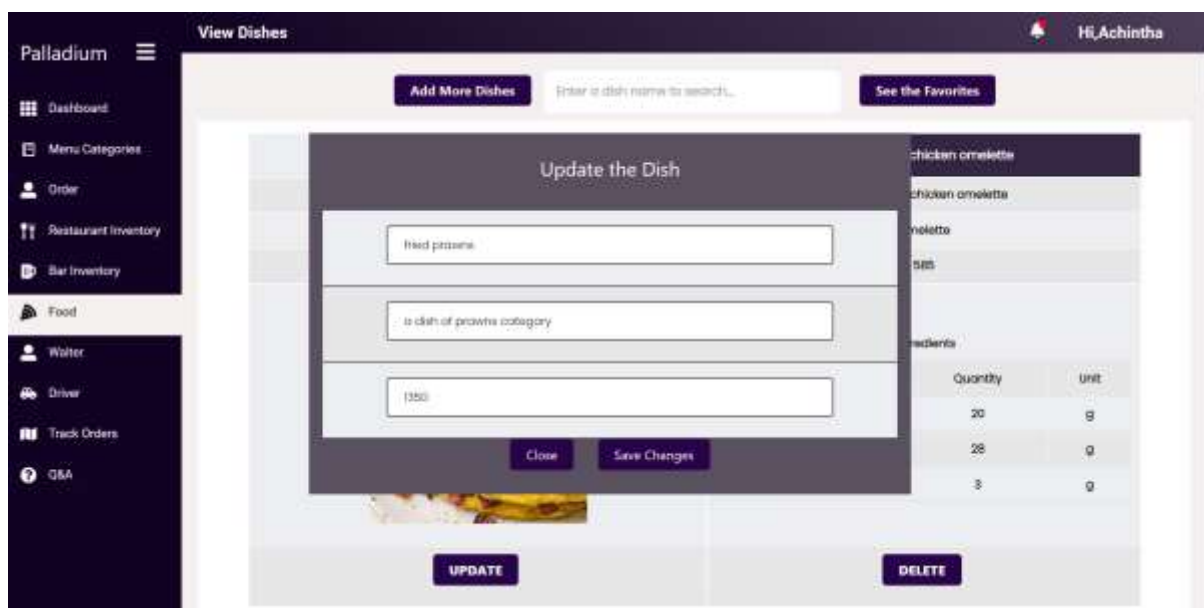


Figure 2.38 Update added dish

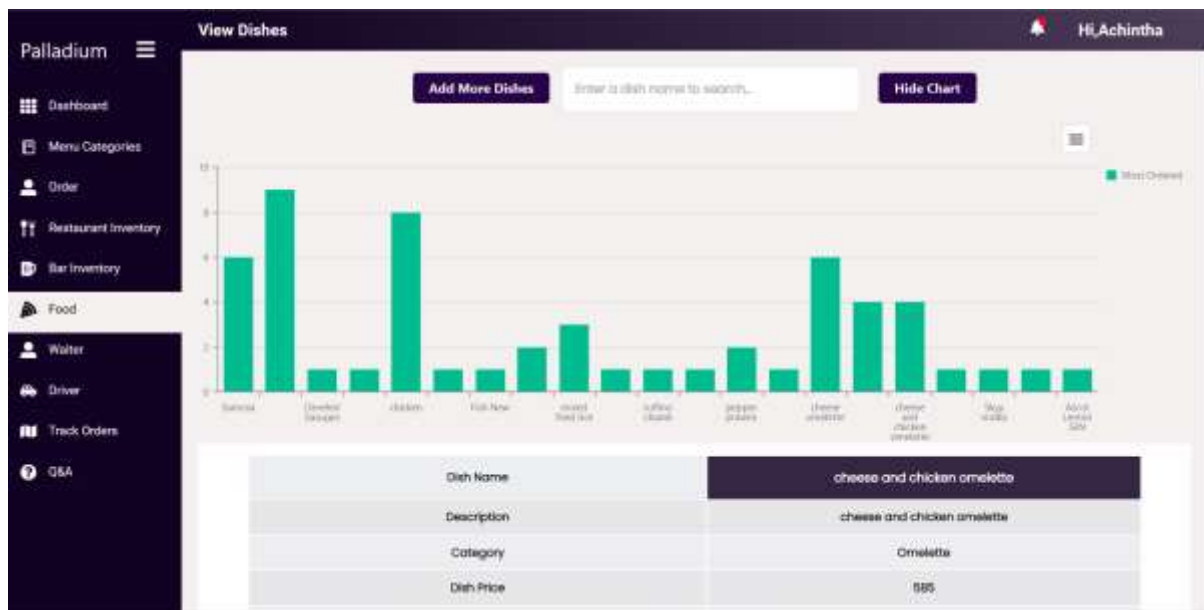


Figure 2.39 Food management Report

Menu

Buttons: Add Category, Category Name, Category Id

Form Fields:

- Category Id:
- Category Name:

Buttons: Add, Cancel

Figure 2.40 Add Menu Category

Category

Buttons: Search Categories..., + New Category

Category Id	Category Name	Action
1	Omelette	Edit Delete
4	Pasta	Edit Delete
5	Sandwiches	Edit Delete
6	Rice	Edit Delete
7	Beef	Edit Delete
8	Seafood	Edit Delete

Figure 2.41 View, Update and Delete Menu Category

Restaurant Inventory/ Add Items Hi,Achintha

Add Items

Item Name: Quantity: Unit Price:

Total Cost: Supplier: Re-order level:

Expiry Date:

Item Id	Item Name	Current Stock	Total Cost	Re-order level
---------	-----------	---------------	------------	----------------

Figure 2.43 Add/Update Restaurant Inventory

Restaurant/ Delete Records Hi,Achintha

Delete the records

Item Id: Item Name: Date:

Time	Buy Date	Quantity	Unit Cost	Buy Cost
22:36	19-5-2023	18	280	5040

Figure 2.42 Delete items in Restaurant Inventory

Restaurant Inventory Hi,Achintha

Stock Summary 18

Item Id	Item Name	Current Stock	Total Cost	Re-order level	Status
17	ymhh	20	2800	10	good
Date + Time Quantity Unit Price Total Cost					
19-5-2023 - 8:21		22	120	2620	
18	eyt	0	0	10	bad
19	tomatoes	90	18000	20	good
20	tomatoes	21	3780	10	good
21	spinach	20	4000	10	good

Figure 2.44 View Restaurant Inventory

Bar Inventory HiAchintha

Add Details

Product Name: Carlsberg	Product Type: Carlsberg	Category: Beer
Unit cost: 500	Quantity: 10	Total cost: 5000
Re-order level: 10	Self price: 0.00	Expiry date: 2024/09/18
photo: 		

Product Code	Product Name	Product Type	Expiry Date	Stock	Re_Order_Level

Figure 2.45 Add/Update Bar Inventory

Bar Inventory HiAchintha

Delete Details

Product Code: Product Name:

Time	Expiry Date	Quantity	Unit cost	Cost	
10:11	2023-06-01	10	2000	20000	☐
10:1	2023-05-24	20	2000	40000	☐

Figure 2.46 Delete Bar Inventory

Bar Inventory HiAchintha

Stock Summary

Product Code: Product Name: Carlsberg

Product Code	Product Name	Product Type	Quantity	Re-Order Level	Total Cost	Status
5	Carlsberg Beer	Beer	10	10	44000	good
1	Red Bull	Energy Drink	20	10	27000	good
4	Wey Water	Water	10	10	20000	good
5	Wey	Energy Drink	20	10	30000	good
5	Wey Water	Red Wine	5	10	1000	low

Code - Time	Quantity	Unit Cost	Total Cost
10:11	10	2000	20000



Figure 2.47 View Bar Inventory

Order ID	amount	Date/time	Waiter Name	type	Customer
30	25395	19-5-2023 - 3:20	hasenberg	Dining	-
31	48280	19-5-2023 - 3:21	-	Takeaway	-
32	50600	19-5-2023 - 3:22	-	Takeaway	-
33	92500	19-5-2023 - 3:23	-	Takeaway	-
34	183500	19-5-2023 - 3:25	-	Takeaway	-
35	475	19-5-2023 - 7:35	-	Takeaway	-
36	275	19-5-2023 - 7:56	-	Takeaway	-
37	585	19-5-2023 - 8:53	-	Delivery	9460e1fa777093b0ca1f033b
38	905	19-5-2023 - 8:58	-	Takeaway	-
39	275	19-5-2023 - 8:59	-	Dining	-
40	575	19-5-2023 - 9:06	-	Takeaway	-
41	16400	19-5-2023 - 9:57	hasenberg	Dining	-

Figure 2.48 Order History

Order

Restaurant | **Bill**

Invoice Number: 40 | Invoice Date: 19/5/2023 - 09:59

Name: _____

Quantity: 1 | Price: _____

Takeaway | Dining | Delivery

Amount: LKR 475

Item	Quantity	Price (LKR)	Amount
cheese omelette	1	475	475

LKR: 475

Printed Bill Preview:

Palladium

Lawson street,
Nuwarasallya

19/5/2023 - 21:27 | NO:47

ITEM	QTY	PRICE	AMOUNT
cheese and chicken omelette	1	585.00	585.00
Ascot Lemon GIN	1	2000.00	2000.00

Net Total: LKR: 2585.00

Please call our hotline
0771413299 for your valued
suggestions and comments.

Figure 2.49 Add Invoice

Palladium

Lawson street,
Nuwarasallya

19/5/2023 - 21:27 | NO:47

ITEM	QTY	PRICE	AMOUNT
cheese and chicken omelette	1	585.00	585.00
Ascot Lemon GIN	1	2000.00	2000.00

Net Total: LKR: 2585.00

Please call our hotline
0771413299 for your valued
suggestions and comments.

Figure 2.50 Printed Bill



Figure 2.52 Report generated for number of messages received per day

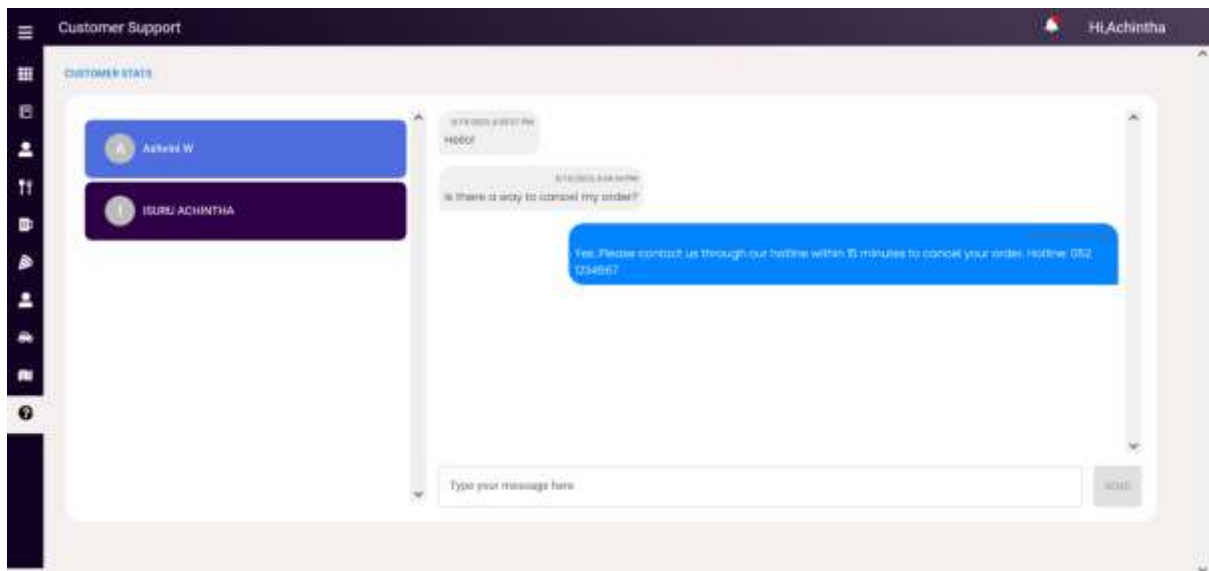


Figure 2.51 Customer – Admin live chat

The screenshot shows the 'Frequently Asked Questions' management interface. At the top, there's a header with a menu icon, the title 'Frequently Asked Questions', and a user profile 'HLAchintha'. Below the header, there's a 'Check Customer Messages' button. The main form area has a 'FAQ Category' section with radio buttons for 'Orders', 'Account', 'Payment', and 'Delivery'. Below this is a 'Question' text input field and an 'Answer' text input field. A '+ Add new FAQ' button is at the bottom of the form. Below the form is a search bar and a table of existing FAQs.

Category	FAQ	Answer	Edit	Delete
Account	Can I change my delivery address after signing up?	Yes. Go to Home > Account > Update your new address and click on update profile.	Edit	Delete

Figure 2.53 Add new FAQ

The screenshot shows the 'Frequently Asked Questions' management interface with an 'Update FAQ' modal form open. The modal has a title 'Update FAQ' and a 'FAQ Category' section with radio buttons for 'Orders', 'Account', 'Payment', and 'Delivery'. Below this is a 'Question' text input field and an 'Answer' text input field. At the bottom of the modal are 'Save' and 'Cancel' buttons. The background shows the same form and table as Figure 2.53, but they are dimmed.

Category	FAQ	Answer	Edit	Delete
Account	Can I change my delivery address after signing up?	Yes. Go to Home > Account > Update your new address and click on update profile.	Edit	Delete
Delivery	Are delivery charges same for everyone?	No. It depends on your location.	Edit	Delete

Figure 2.54 Update FAQs

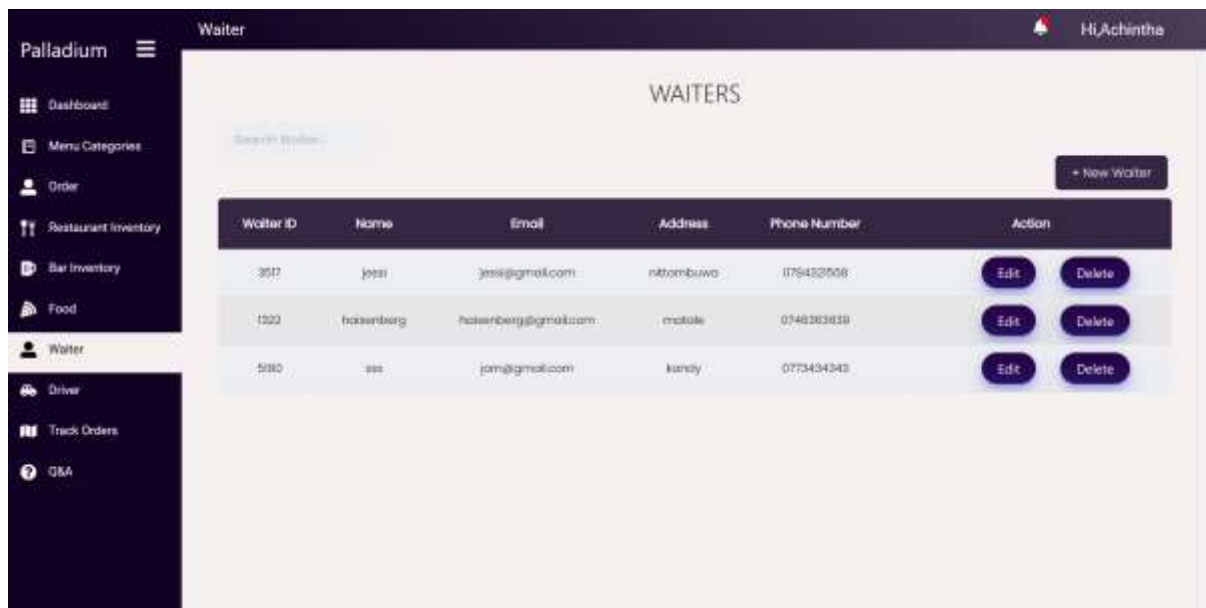


Figure 2.55 View/Update/Delete Waiters

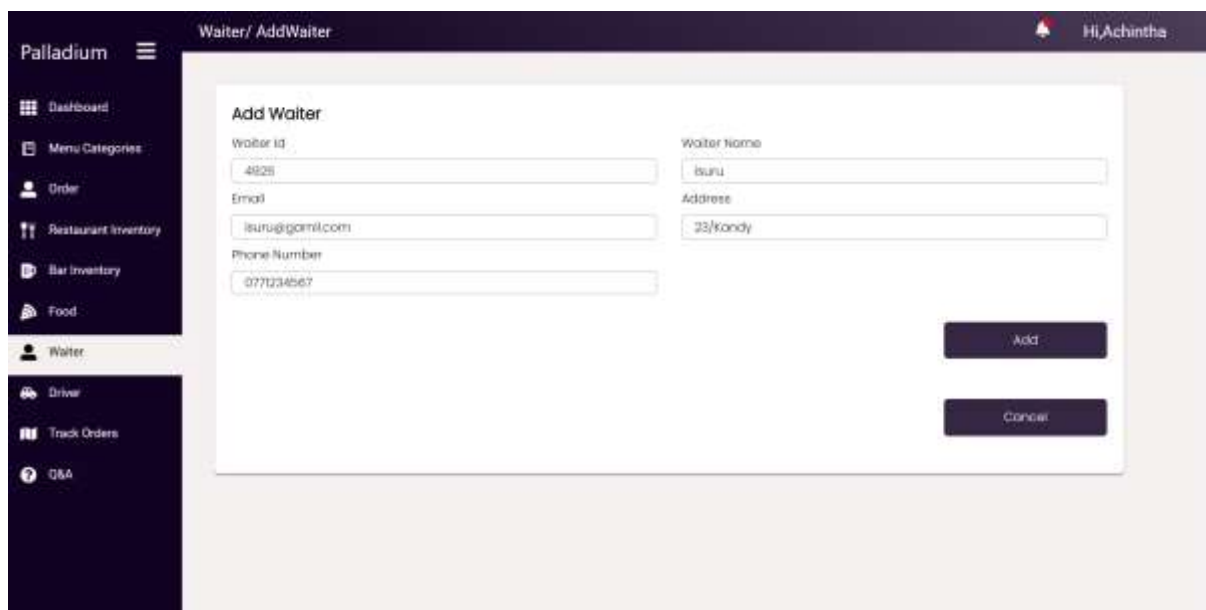


Figure 2.56 Add new Waiters

Palladium Driver/ Add Driver HiAchintha

New Driver

Driver ID: 3058

Driver Name: kamal

Email: kamal@gmail.com

Address: Kandy

Phone Number: 977234567

Add

Cancel

Figure 2.57 Add new Drivers

Palladium Driver HiAchintha

DRIVERS

Search Drivers...

[+ New Driver](#)

Driver ID	Name	Email	Address	Phone Number	Action
1825	kkj	ghg@gmail.com	tytygh	9767676767	Edit Delete
1896	asd	asd@gmail.com	121/nuwara	987001	Edit Delete
3036	afrah	afrah@gmail.com	kandy34	9772345646	Edit Delete

Figure 2.58 View/Update/Delete Drivers

2.3 Implementation

2.3.1 Tools and Technologies used for the Development.

In our Information Technology Project, we utilized UML tools and techniques along with a database modeling program, MongoDB. Our tasks were firstly separated into sections according to the functionalities among each member with equal scope. To make models and graphics for our projects, we used Mock flow and Draw.io software's. Every single member of our team kept their promises and worked on their given section. Microsoft PowerPoint was used to construct the proposal presentation. Microsoft Word was utilized to complete our task in preparing the proposal presentation. The Microsoft Office suite has thus far proven to be of great assistance to us in completing our project proposal planning and other utilities. Each team member individually displayed their skills in Word, and then they all merged their efforts to create a single word document.

A useful free outlining tool, Draw.io provides its users with powerful collaborative tools. It has a variety of shape libraries, each of which has many graphic elements. We were able to use it to make our framework outline chart as a result. These are the techniques and technological developments that we have utilized.

To complete our project on schedule we drew on a range of resources. To achieve our objectives, we used JavaScript and Node.js as programming languages. It is a JavaScript runtime environment that enables the execution of JavaScript code outside of a web browser. The JavaScript engine in version 8 serves as the foundation for the entire environment. Building high-performance web applications using Node.js is a real possibility. We used Visual Studio Code as our main integrated development environment.

We built our application using Express as the back-end framework and React as the front-end framework. Express is a NodeJS web application framework that keeps multi-platform and web applications simple while incorporating features. It explains how worker-side applications for the web and portable devices are necessary. The MERN stack's backend is made up of Express and the MongoDB data store, with React acting as the frontend framework. React is a JavaScript library for designing adaptable and efficient user interfaces. It builds intricate user interfaces out of discrete, small pieces of code called "components." It is utilized to build single-page web applications.

Working with other members was made easier by using GitHub. Team members can easily edit files and merge their changes with the project's master branch. It shortens the time we need for integration.

By using these technologies and software's, planning, and developing the said system has been efficient, effective and less time consuming with ease of work.

Software used	Purpose
Visual Studio Code	Code editor
MongoDB	Cross-platform document-oriented database
Google chrome	Web browser for viewing
Adobe Illustrator	Logo designing
React	Frontend development
Node.js	Backend server environment

Table 2-1 Development tools and technologies

2.3.2 Conventions and validations followed.

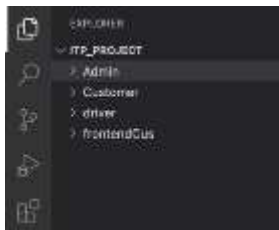


Figure 2.60 Folder structure



Figure 2.63 Folder structure - Admin

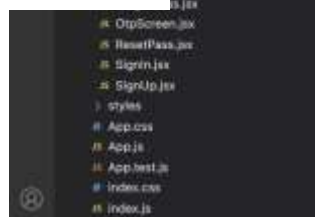


Figure 2.62 Folder structure - Customer

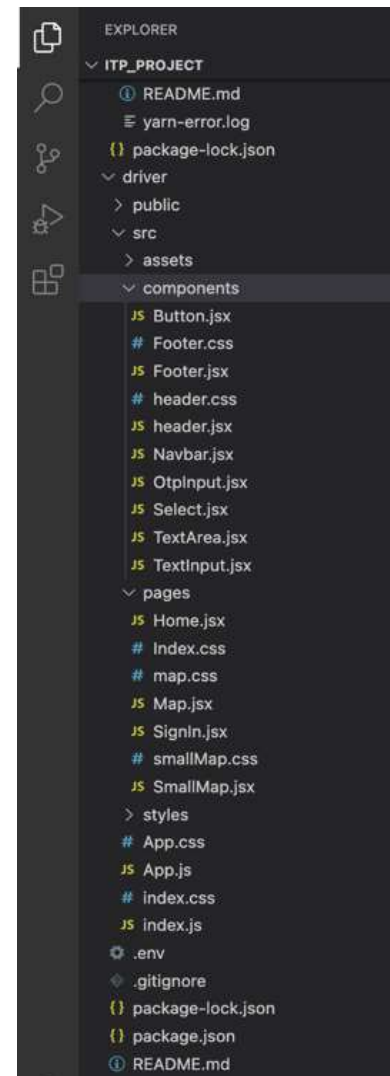


Figure 2.61 Folder structure - Driver

Screenshots of some validations

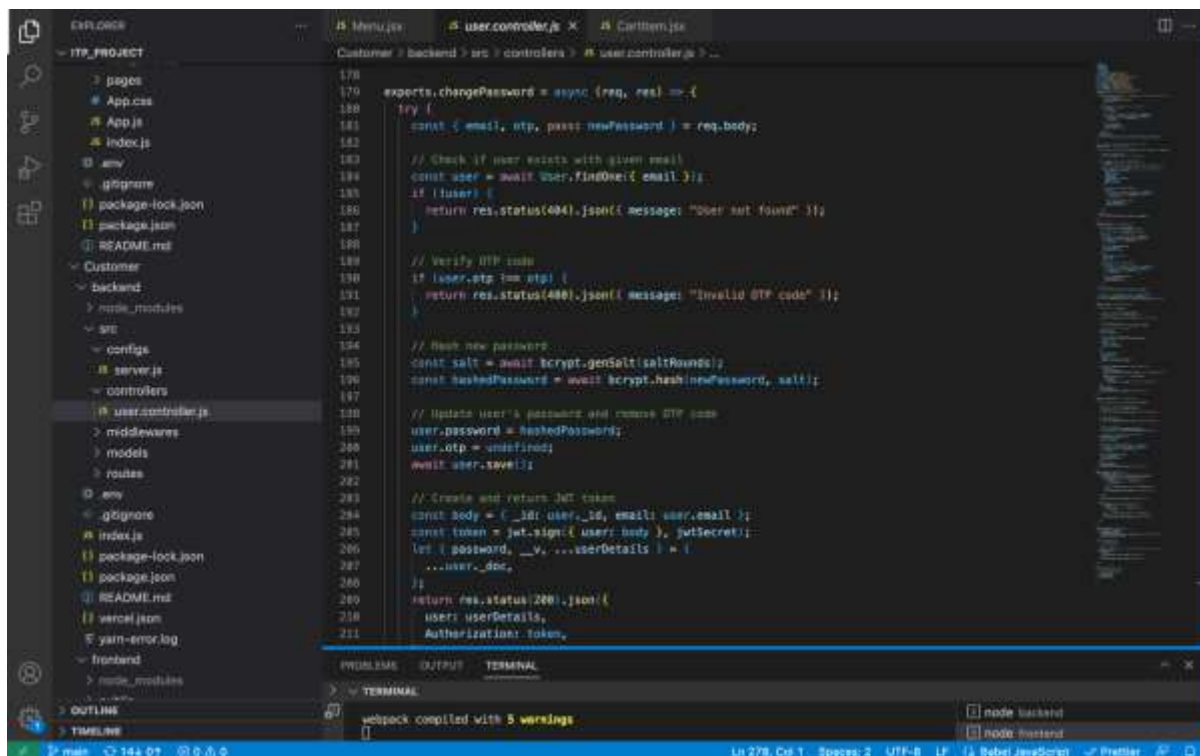


Figure 2.64 User management validations

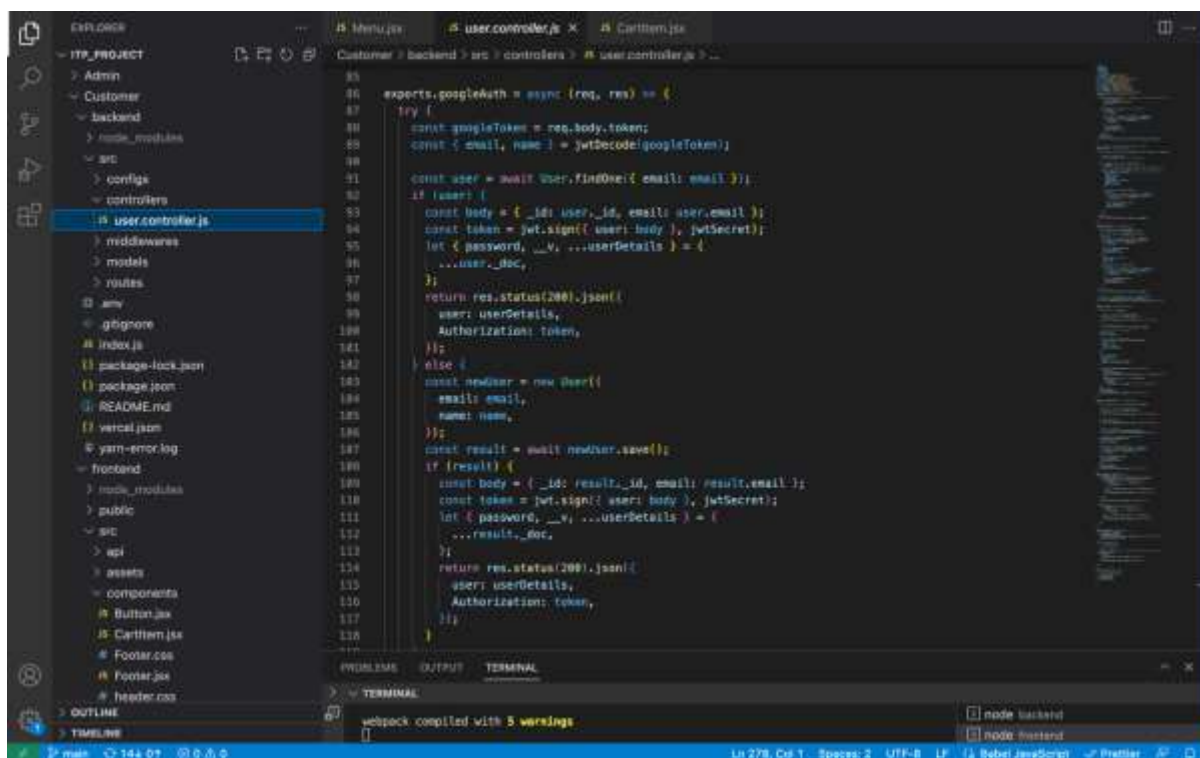


Figure 2.65 User management validations using Google auth

```

<Button
  // style={{ backgroundColor: "#01BC90", color: "black" }}
  type="submit"
  id="print"
>
  Print Bill
</Button>
)}
onBeforePrint={() => {
  if (total <= 0) {
    toast.error("Please add food");
    throw new Error("Printing is not allowed!");
  } else if (type == "Dining" && table == "") {
    toast.error("Please select table ");
    throw new Error("Printing is not allowed!");
  } else if (
    staytus === "0" &&
    type === "Delivery" &&
    !address
  ) {
    toast.error("Please enter the address");
    throw new Error("Printing is not allowed!");
  } else if (type == "") {
    toast.error("Please set order type");
    throw new Error("Printing is not allowed!");
  } else {
    setPrint(1);
  }
}}
content={() => componentRef.current}

```

Figure 2.66 Form validations

```

<td>
  <Autocomplete
    onChange={(e) => setAddress(e.target.value)}
  >
    <input
      type="text"
      hidden={true}
      id="locationInputs"
      style={{ padding: "1rem 1rem 1rem 1rem" }}
      placeholder="Enter your drop point..."
      onChange={(e) => setAddress(e.target.value)}
      onClick={(e) => setAddress(e.target.value)}
      onMouseMove={(e) =>
        setAddress(e.target.value)
      }
      onMouseUp={(e) => setAddress(e.target.value)}
      value={address}
    />
  </Autocomplete>
</td>

```

Figure 2.67 Location picker

```

const s3 = new AWS.S3();
const [file, setFile] = useState(null);
const validFileTypes = ["image/jpg", "image/jpeg", "image/png"];
AWS.config.update({
  accessKeyId: "AKIAV3TWWOPNV5Z3UJ6X",
  secretAccessKey: "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",
  dirName: "images",
  region: "ap-south-1",
  signatureVersion: "v4",
});
const handleFileSelect = (e) => {
  const file_ = e.target.files[0];

  if (!validFileTypes.find((type) => type === file_.type)) {
    toast.error(
      "Enter the dish Name first. Then select an JPG/PNG file type."
    );
    return;
  } else {
    setFile(e.target.files[0]);
  }
};

```

Figure 2.68 image upload using AWS S3 bucket

```

<Chart
  id="chart"
  // title="<b>Sales Report</b>"
  dataSource={grossProductData}
  onPointClick={onPointClick}
>
  <CommonSeriesSettings
    argumentField="state"
    type="bar"
    hoverMode="allArgumentPoints"
    selectionMode="allArgumentPoints"
  ></CommonSeriesSettings>
  <Series
    argumentField="state"
    valueField="Takeaway"
    name="Takeaway"
    color="#01BC90"
  />
  <Series valueField="Dining" name="Dining" color="#00A0AC" />
  <Series valueField="Delivery" name="Delivery" color="#0083BB" />
  <Legend verticalAlignment="bottom" horizontalAlignment="center"></Legend>
  <Export enabled={true} />
</Chart>

```

Figure 2.69 generating bar chart

3 Testing and Evaluation

3.1 Testing Methodologies

To ensure the functionality, reliability, and usability of the new restaurant management system, a comprehensive testing approach was implemented. The following testing methodologies were employed during the testing phase:

3.1.1 Unit Testing

Unit testing was conducted to verify the functionality of individual system components, modules, and functions. It involved testing each unit in isolation to ensure that it operates correctly and produces the expected output.

3.1.2 Integration Testing

Integration testing was performed to examine the interaction and compatibility of different system modules. It aimed to identify any issues or inconsistencies that may arise when the modules are integrated into the complete system.

3.1.3 System Testing

System testing was conducted to evaluate the overall behavior and performance of the restaurant management system. This involved testing the system as a whole to ensure that all modules work together seamlessly and meet the specified requirements.

3.1.4 User Acceptance Testing

User acceptance testing was carried within the team as it cannot be tested with the involvement of the client as the system is developed only for educational purpose. The purpose was to evaluate the system's usability, functionality, and user satisfaction.

3.2 Testing Results

The testing phase yielded positive results, demonstrating the effectiveness and reliability of the new restaurant management system. The following are the key findings from the testing process:

3.2.1 Functionality Testing

All system functionalities, including order placement, inventory management, and reporting, were thoroughly tested. The system accurately processed customer orders, generated reports, and maintained inventory records as expected.

3.2.2 Performance Testing

Performance testing was conducted to assess the system's responsiveness and scalability. The system performed efficiently demonstrating its ability to handle a large number of concurrent process transactions swiftly.

3.2.3 Usability Testing

Usability testing was carried out to evaluate the system's user-friendliness and intuitiveness. Feedback from the client indicated that the system was easy to navigate, and the user interface facilitated smooth and efficient order management.

3.2.4 Error Handling and Exception Testing

Various error scenarios and exceptions were intentionally triggered to assess the system's error-handling capabilities. The system demonstrated robust error handling, providing appropriate error messages and gracefully recovering from errors without compromising data integrity.

3.3 Evaluation

Based on the testing results and user feedback, the new restaurant management system has proven to be a significant improvement over the client's previous system. The system successfully addressed the identified problems and met the specified requirements. The following are the key evaluation points:

- **Improved Operational Efficiency**
The new system streamlined order management, inventory control, and reporting processes, resulting in increased operational efficiency. Daily reports were generated accurately and in a timely manner, providing the client with valuable insights into sales and inventory.
- **Enhanced Customer Experience**
With the online ordering system and improved order management, customers experienced greater convenience and faster service. The system enabled seamless ordering from various channels and improved accuracy in order fulfillment.
- **Minimized Fraud and Wastage**
The new inventory management module significantly reduced instances of fraud and wastage. The system provided real-time inventory tracking, ensuring better control over stock levels and minimizing inventory-related losses.
- **User Satisfaction**
User acceptance testing feedback indicated high levels of satisfaction in the perspective of the restaurant staff. The system's user-friendly interface and intuitive workflows minimized the need for additional training, allowing staff to adapt quickly and perform tasks efficiently.

In conclusion, the testing phase demonstrated the effectiveness, reliability, and usability of the new restaurant management system. The system's functionalities, performance, error handling, and user satisfaction met the desired objectives and successfully addressed the challenges identified

3.1 Test cases

3.1.1 User Management

Testing function	User management
Test case id	01
Test priority	High
Test description	User will get signed in into to the system
Pre condition	User must have a registered account
Post condition	Login authentications approve and user will be logged in to his/her user profile
Test steps	1.visit the website 2.Enter username, password 3.Allow access to the system

Table 3-1 Test case designed by: IT21367458 Siyambalapitiya H.N

Test Id	Test input	Result	Expected output	Actual output
001	Login to the system with correct email with incorrect password Email – abc@gmail.com Password -	fail	Customer login successfully to the system	System shows an error message
	Login to the system with correct email and correct password Email – abc@gmail.com Password - 1234	pass	Customer login successfully to the system	Allow access to the system
002	Customer didn't filled the all input fields	fail	Update user details	System show a message to fill all fields
	Customer fill the all inputs with correct details	pass	Update user details	Update user account correctly

Table 3-2 Test case designed by: IT21367458 Siyambalapitiya H.N

3.1.2 Food management

Testing function	Food management
Test case id	02
Test priority	Medium
Test description	admin add/update or delete a dish item
Pre condition	User must log in to the system as an admin
Post condition	new dish must add/update in the system
Test steps	1.Admin choose to add a new dish 2.Admin enters dish details 3.System validates details 4.System stores details in the database 5.Database returns data insertion progress 6.System displays dishes in the dish management section.

Table 3-3 Test case designed by: IT21209352 Bandara D.M.K.C

Test id	Test input	Result	Expected output	Actual output
003	Check the image type other than jpg or png	fail	Add dish to the system	System shows an error message
	Add dish without filling any form details	fail	Add dish to the system	System shows an error message
	Fill all the required details with an image Dish name – fried rice Description – fried rice Price – 500/= Select dish category – rice Unit – g Quantity – 250 Image – rice.jpg	pass	Add dish to the system	Adding dish correctly

Table 3-4 Test case designed by: IT21209352 Bandara D.M.K.C

Testing function	Menu category management
Test case id	03
Test priority	Medium
Test description	Categorize the dishes
Pre condition	Admin logs in with high priority credentials
Post condition	Menu items are grouped and displayed accordingly
Test steps	1.Admin enters the application 2.Admin signs in 3.Admin goes to the category tab 4.Admin selects to add a new category 5.A form is displayed 6.Admin fills out the form with the details 7.Admin submits the form 8.Database is updated 9.Available categories are displayed

3.1.3 Menu category management

Table 3-5 Test case designed by: IT21294952 Aluthge I.N

Test id	Test input	Result	Expected output	Actual output
004	Add category without filling details	fail	Add category to system	System shows an error message
	Fill required details correctly Category id – 1001 Category name - chicken	pass	Add category to the system	Added category correctly
005	Click cart without adding any dishes	fail	Add dishes to the cart	System show message to add dishes
	Adding dishes with correct quantity and price	pass	Add dishes to the cart	Added dishes correctly

Table 3-6 Test case designed by: IT21294952 Aluthge I.N

3.1.4 Order management

Testing function	Order management
Test case id	04
Test priority	High
Test description	Cashier manage the order
Pre condition	Cashier login to system
Post condition	Print the invoice
Test steps	1.Casher login to system 2.Navigate to order page 3.Choose order type 4.Select Dish/ bottles 5.Select quantity 6.Choose order method 7.Add item for invoice 8.Casher can update/delete item 9.View success message

Table 3-7 Test case designed by: IT21281464 Wijethunga I.A

Test id	Test input	Result	Expected output	Actual output
006	Select restaurant delivery without filling any form details	fail	Order added in restaurant based on selected order type	Order not success
	Select take away order with filling all details Name – fried rice Quantity – 3 Price – 1500 Type – take Away	pass	Order added in restaurant based on selected order type	Place an order successfully
007	Make delivery order without giving address	fail	Place a delivery order	Order cancelled

Table 3-8 Test case designed by: IT21281464 Wijethunga I.A

3.1.5 FAQ management

Testing function	FAQ management
Test case id	05
Test priority	Medium
Test description	Admin adding frequently asked questions and relevant answers/solutions
Pre condition	Admin accessing the admin dashboard
Post condition	Customer can view the FAQs added by the admin, through the web application
Test steps	1.Admin enters login details and logs in to the website 2.Admin is redirected to customer dashboard 3.Admin clicks on “Add new FAQ” button 4.Admin chooses FAQ category 5.Admin adds the new FAQ and the relevant answer 6.Newly added FAQ is displayed in the admin dashboard and the FAQ tab of the web application 7.Admin can edit or remove already added FAQs

Table 3-9 Test case designed by: IT21276446 Wegodapola A.R

Test id	Test input	result	Expected output	Actual output
008	Test input add new FAQ and answer without choosing category type	fail	Add FAQ to the system	System show an error message
	Add new FAQ without answer Question – Can we order online? Answer -	fail	Add FAQ to the system	System show an error message
	Add new FAQ and answer with choosing category type FAQ category – payment Question – how can I pay? Answer – you can pay on delivery	pass	Add FAQ to the system	Added question correctly

Table 3-10 Test case designed by: IT21276446 Wegodapola A.R

3.1.6 Restaurant inventory management

Testing function	Restaurant inventory management
Test case id	06
Test priority	High
Test description	Inventory manager adding new items to the system
Pre condition	Admin must login to the system
Post condition	Update inventory table frequently
Test steps	1.Manager login to the system 2.Navigate to the restaurant inventory page 3.Inventory page is displayed 4.Manager selects “add items” and show the add form 5.Fill the add form 6.submit the form 7.view status 8.Manger can delete inventory records

Test id	Test input	result	Expected output	Actual output
009	Adding items without fill required fields	fail	Add Items to the inventory	System show an error message
	Adding items with incorrect input types	fail	Add Items to the inventory	System show an error message
	Fill all fields and add item Item id – 001 Item name – chicken Buy date – 2023/03/20 Quantity – 23 Unit – Kg Unit price – 680 Total cost – 23000 Supplier – cc farm Re-order level - 10	pass	Add Items to the inventory	Added item successfully

Table 3-11 Test case designed by: IT21373770 Dissanayake D.M.U.I

Table 3-12 Test case designed by: IT21373770 Dissanayake D.M.U.I

3.1.7 Bar inventory management

Testing function	Bar inventory management
Test case id	07
Test priority	High
Test description	manage bar stock levels
Pre condition	Admin login to system
Post condition	Inventory must update frequently
Test steps	1.inventory manager login to system 2.inventory manager click bar inventory page 3.inventory manager direct to bar inventory page 4.inventory manager check inventory 5.inventory manager redirect to add form 6.inventory manager add inventory 7.Validate inventory 8.Receive notification

Table 3-13 Test case designed by: IT21239434 Punchihewa P.H.K.C

Test id	Test input	result	Expected output	Actual output
010	Input image type other than jpg or png	fail	Add product to the inventory	System show an error message
	Adding items without fill required fields	fail	Add product to the inventory	System show an error message
	Fill all required details with adding image Product code – B002 Item name – vodka Category - hard Quantity – 5 Unit price – 1800 Buy date – 2023/03/20 Total cost – 9000 Re-order level - 15 Sell price – 2400 Expire date - 2024/04/10 Image – a1.jpg	pass	Add Product to the inventory	Added product successfully

Table 3-14 Test case designed by: IT21239434 Punchihewa P.H.K.C

3.1.8 Waiter management

Testing function	Waiter management
Test case id	08
Test priority	Medium
Test description	Add new waiter
Pre condition	Admin must have login to the system
Post condition	Waiters must manage
Test steps	1.admin login to system 2.admin click waiter page 3.admin direct to waiter page 4.admin direct to add form 5.admin add waiter 6.Validate information 7.Admin update waiter details 8.Admin remove waiters

Table 3-15 Test case designed by: IT21296796 S.M.Shabry

Test id	Test input	result	Expected output	Actual output
011	Add waiter without filling details	fail	Add waiters to the system	System show an error message
	Fill fields with incorrect email or phone number	fail	Add waiters to the system	System show an error message
	Add waiter with correct details Waiter id – W002 Waiter name – Kamal Email – abc@gmail.com Address – Kandy Phone number - 077123456	pass	Add waiters to the system	Waiter adding to the system successfully

Table 3-16 Test case designed by: IT21296796 S.M.Shabry

3.1.9 Driver management

Testing function	Driver management
Test case id	09
Test priority	medium
Test description	Add new driver
Pre condition	Admin login to system
Post condition	drivers must manage
Test steps	1.admin login to system 2.admin click driver page 3.admin direct to driver page 4.admin direct to add form 5.admin add drivers 6.Validate information 7.Admin update driver details 8.Admin remove driver

Table 3-17 Test case designed by: IT21296550 M.I.Ahlan

Test id	Test input	result	Expected output	Actual output
012	Login with incorrect email or password	fail	Login to the system as a driver	System show an error message
013	Add driver without filling details	fail	Add driver to the system	System show an error message
	Add waiter with correct details Driver id – D003 Driver name – Kamal Email – abc@gmail.com Address – Kandy Phone number - 077123456	pass	Add driver to the system	Driver adding to the system successfully

Table 3-18 Test case designed by: IT21296550 M.I.Ahlan

1. Conclusion

In conclusion, the Restaurant Management System has successfully achieved its original objectives and goals. Throughout the project, we have addressed the various challenges faced by our client and developed a comprehensive solution to streamline their operations and enhance customer satisfaction.

The system effectively addresses the limitations of the previous desktop application used by our client. It provides features such as online ordering, dine-in management, delivery system, inventory management, and waiter and driver management. These functionalities have greatly improved the overall efficiency and effectiveness of the restaurant's operations.

However, it is important to acknowledge the weaknesses and limitations of our proposed technique. One limitation is the dependency on stable internet connectivity for seamless online ordering and real-time updates. In areas with unreliable internet access, the system's performance may be affected. To mitigate this limitation, it is recommended to have backup solutions in place, such as offline mode functionality or alternative communication channels for order placement.

Another limitation is the initial learning curve for staff members in adapting to the new system. While efforts have been made to ensure user-friendliness, additional training and support may be required to help employees fully utilize the system's features and maximize its benefits. Providing comprehensive training sessions and user manuals can help overcome this limitation.

To address these weaknesses, we recommend ongoing system maintenance and updates to ensure compatibility with evolving technologies and address any identified bugs or issues. Regular feedback from users and customers should be solicited to identify areas for improvement and implement suggested solutions.

In conclusion, the Restaurant Management System has proven to be a valuable tool for our client, addressing their specific requirements and improving their overall operations. By continually refining and enhancing the system based on user feedback, we can ensure its long-term success and assist our client in staying ahead in the competitive restaurant industry.

1.1 References

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Appendices

GitHub Repository:

[GitHub Repository - KDY_2023_WD_G1 - Restaurant Management System](#)